

AI-DRIVEN BLOCKCHAIN SYSTEM FOR PROACTIVE CYBER THREAT DETECTION

A PROJECT REPORT

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BONAFIDE CERTIFICATE

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We consider it our privilege to express gratitude to a few people who have guided and inspired us in the completion of this project. The journey of developing this project — an AI-Driven Blockchain System for Proactive Cyber Threat Detection — has been both intellectually stimulating and highly rewarding.

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ABSTRACT

The escalating sophistication of modern cyber threats — particularly zero-day exploits and Advanced Persistent Threats (APTs) — has rendered conventional signature-based security architectures increasingly inadequate. These traditional systems operate reactively, depending on pre-existing signatures to identify malicious activity, leaving organizations entirely vulnerable to novel, previously unseen attack vectors. Furthermore, the centralized nature of most security log management systems introduces a critical single point of failure, enabling attackers to manipulate or delete forensic evidence after a successful intrusion.

This project report presents ProSec-Block (Proactive Security via Blockchain), a novel hybrid framework that integrates the predictive power of Artificial Intelligence (AI) with the trust and immutability guarantees of Blockchain technology to establish a proactive, transparent, and collaborative cyber defense ecosystem. The core of the system is a deep learning-based anomaly detection engine constructed upon an LSTM Autoencoder architecture. This engine is trained exclusively on benign network traffic, enabling it to identify deviations from normal behavior as potential threats — even those lacking any known attack signature.

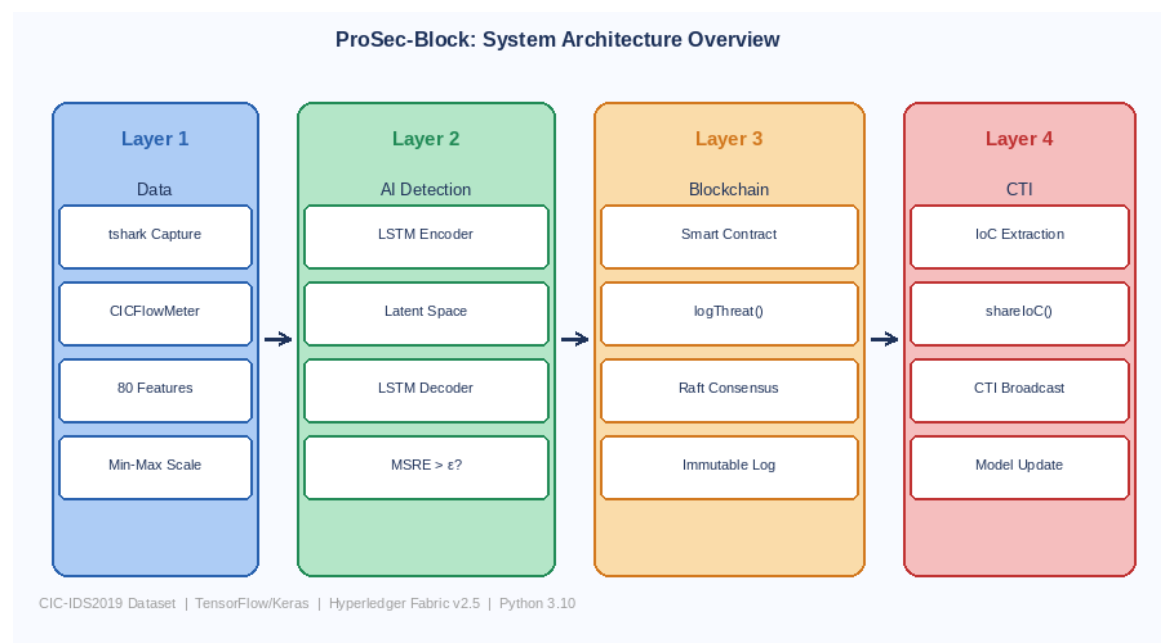
Upon detecting an anomaly, the system automatically invokes a smart contract on a permissioned Hyperledger Fabric blockchain to record the threat metadata onto an immutable distributed ledger, eliminating any possibility of log tampering. The same blockchain infrastructure serves as a medium for decentralized Collaborative Threat Intelligence (CTI) sharing, with verified Indicators of Compromise (IoCs) continuously feeding back into the AI model to improve detection capabilities over time.

Experimental evaluation on the CIC-IDS2019 dataset demonstrates an overall detection accuracy of 98.2%, precision of 97.9%, recall of 98.4%, and F1-Score of 98.1%, with a False Positive Rate of only 1.5%. The blockchain sustains a throughput of 450 transactions per second with average consensus latency of 2.1 seconds. The integrated pipeline achieves a 40% reduction in end-to-end incident response time compared to conventional architectures, validating the system's potential as a foundation for next-generation decentralized Security Operations Centers (SOCs).

Keywords: Blockchain Security, Artificial Intelligence, Deep Learning, Proactive Threat Detection, Intrusion Detection System (IDS), Immutable Ledger, Threat Intelligence Sharing, Smart Contracts, Zero-Day Attacks, LSTM Autoencoder, Hyperledger Fabric.

GRAPHICAL ABSTRACT

The figure below presents a visual overview of the ProSec-Block system, illustrating the end-to-end flow from raw network traffic, through the AI detection engine, into the blockchain logging layer, and finally into the collaborative threat intelligence sharing module with a feedback loop back to the AI model.



Graphical Abstract: High-Level Visual Overview of the ProSec-Block AI-Blockchain Cyber Threat Detection System

The diagram depicts the four primary stages of the ProSec-Block framework. In Stage 1 (Data Acquisition), raw network traffic packets from various endpoints are captured using tshark and preprocessed into 80 flow-based feature vectors. In Stage 2 (AI Detection), these features are fed into the LSTM Autoencoder in batches of 100 packets, and sequences with reconstruction error exceeding threshold ϵ are flagged as potential anomalies. In Stage 3 (Blockchain Integration), a smart contract on Hyperledger Fabric is invoked to log the threat metadata immutably. Finally, in Stage 4 (CTI Sharing), verified Indicators of Compromise (IoCs) are broadcast across consortium nodes and fed back into the AI model for continuous learning.

ABBREVIATIONS AND SYMBOLS

List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
APT	Advanced Persistent Threat
BC	Blockchain
CTI	Collaborative / Cyber Threat Intelligence
DDoS	Distributed Denial of Service
DL	Deep Learning
DoS	Denial of Service
FPR	False Positive Rate
GNN	Graph Neural Network
IDS	Intrusion Detection System
IoC	Indicator of Compromise
IoT	Internet of Things
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MISP	Malware Information Sharing Platform
ML	Machine Learning
MSRE	Mean Squared Reconstruction Error
NIST	National Institute of Standards and Technology
RNN	Recurrent Neural Network
SHA	Secure Hash Algorithm
SOC	Security Operations Center
SVM	Support Vector Machine
TPS	Transactions Per Second
XAI	Explainable Artificial Intelligence

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CHAPTER 1 INTRODUCTION

1.1 Background and Motivation

We live in an age where nearly every facet of modern society — from banking and healthcare to national defence and critical infrastructure — is intricately dependent on interconnected digital systems. This profound digitalization has unlocked unprecedented levels of productivity and global connectivity. However, it has simultaneously created an enormously expanded attack surface that malicious actors exploit with growing sophistication [1]. Cyber threats are no longer the domain of lone hackers; they have evolved into state-sponsored espionage campaigns, organized criminal syndicates deploying Ransomware-as-a-Service (RaaS) platforms, and advanced persistent actors capable of remaining undetected within a victim's network for months or even years [20].

The fundamental challenge of modern cybersecurity lies in the asymmetric nature of attack and defence. An attacker needs to find and exploit only one vulnerability; a defender must protect against all possible avenues simultaneously. Traditional security architectures — built around firewalls, antivirus software, and signature-based Intrusion Detection Systems (IDS) — were designed for a threat landscape that no longer exists [16]. These systems operate on a reactive paradigm: they compare observed network traffic against a database of known attack signatures and raise alerts only when a match is found. While effective against well-catalogued threats, they are fundamentally blind to zero-day exploits and novel malware variants that bear no resemblance to existing signatures [14].

Figure 1.1 below provides a high-level illustration of the ProSec-Block architecture that this project proposes as a solution to these challenges.

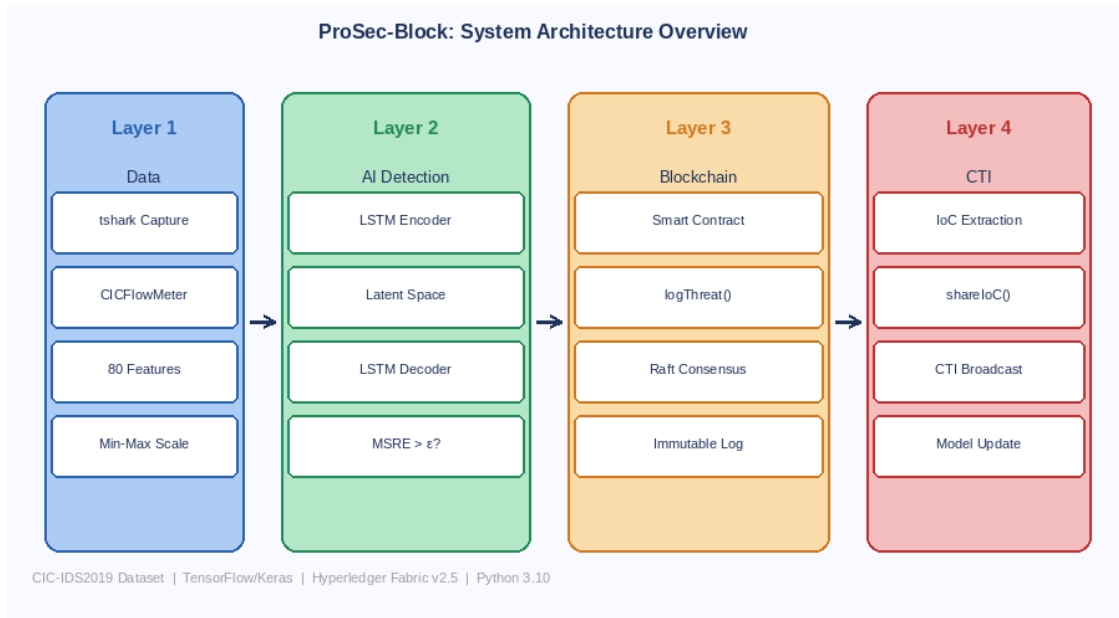


Figure 1.1: ProSec-Block System Architecture — Four-Layer Hybrid AI-Blockchain Framework

1.2 Identification of the Problem

1.2.1 The Zero-Day Vulnerability Crisis

A zero-day vulnerability refers to a previously unknown security flaw in software or hardware that has not yet been discovered or disclosed to the vendor or the public. Because no patch or fix exists at the time of exploitation, systems remain inherently vulnerable. More importantly, since traditional intrusion detection systems rely on known signatures or patterns of past attacks, they are effectively blind to zero-day exploits—there is simply no predefined signature to match against [14][16].

This creates a critical security gap known as the “window of exposure,” which is the period between when an attacker first discovers and exploits the vulnerability and when a patch is eventually released and deployed. This window can last anywhere from a few days to several months—or even over a year—depending on the complexity of the vulnerability and the responsiveness of the vendor. During this time, attackers can operate with minimal resistance, often evading detection entirely while exploiting systems at scale.

A well-known example of the devastating impact of such vulnerabilities is the WannaCry ransomware attack. This attack leveraged the EternalBlue exploit targeting a vulnerability in the Windows SMB protocol, allowing it to spread rapidly across networks worldwide. The result was widespread disruption across critical sectors, including healthcare, transportation, and government services, highlighting how dangerous zero-day or unpatched vulnerabilities can be when weaponized.

This reality underscores the limitations of signature-based detection approaches and reinforces the need for more advanced techniques, such as anomaly-based detection powered by machine learning [2][23]. By focusing on deviations from normal behavior rather than known attack patterns, such systems offer a more effective defense against previously unseen threats, including zero-day exploits [2].

1.2.2 The Log Integrity and Tampering Problem

When a security incident occurs, forensic investigators depend heavily on system and network logs to reconstruct the exact sequence of events leading up to and following the breach [16]. These logs serve as the primary source of truth for understanding attacker behavior, identifying compromised assets, and determining the scope of the incident. However, in traditional architectures, logs are typically stored in centralized databases or logging servers that reside within the same network environment being monitored.

This design introduces a critical vulnerability. If an attacker successfully compromises the network and gains elevated privileges—such as administrative or root access—they can directly interact with the logging infrastructure. With sufficient access, it becomes trivial to modify, delete, or even fabricate log entries. By doing so, attackers can erase evidence of their activities, hide indicators of compromise, and mislead investigators attempting to trace the attack path [20].

Such actions are not theoretical but are well-documented tactics used by advanced adversaries. Log tampering and deletion are standard components of the operational playbook in sophisticated attacks, particularly in Advanced Persistent Threat (APT) campaigns [20]. By removing or altering logs, attackers significantly hinder forensic analysis, delay incident response, and reduce the likelihood of detection or attribution.

This fundamental weakness highlights the need for tamper-resistant logging mechanisms. Without strong guarantees of log integrity, even the most advanced detection systems lose their effectiveness in post-incident analysis. Ensuring that logs remain immutable, verifiable, and resistant to unauthorized modification is therefore essential for building a trustworthy and resilient cybersecurity infrastructure [11].

1.2.3 The Problem of Siloed Threat Intelligence

Effective defence against sophisticated cyber threats requires collective intelligence. When one organization identifies a new attack vector — an Indicator of Compromise (IoC) — that information is potentially valuable to hundreds of other organizations facing the same threat. However, the sharing of threat intelligence remains fragmented in practice.

Organizations are reluctant to share threat data due to concerns about competitive disadvantage and reputational damage. Even centralized platforms like MISIP represent a single point of trust and failure — if the central server is compromised, the threat intelligence itself can be poisoned [9][17].

1.3 Problem Statement

Existing cybersecurity systems are largely reactive in nature, relying on known attack signatures and predefined rules to identify threats. This makes them inherently limited when facing modern attack vectors such as zero-day exploits, where no prior signature exists [14]. In addition, most traditional architectures are centralized, meaning that both detection logic and log storage are concentrated in a single or limited set of systems. This centralization creates critical vulnerabilities, including single points of failure, susceptibility to log tampering, and the risk of compromised or poisoned threat intelligence repositories [17].

These limitations highlight a clear and urgent need for a new approach to cybersecurity—one that is proactive, decentralized, and trust-oriented by design. First, the system must be capable of detecting previously unknown threats through behavioral analysis rather than relying solely on historical attack patterns. By leveraging advanced machine learning techniques, such as anomaly detection models, the system can identify deviations from normal network behavior, enabling it to detect zero-day attacks and novel intrusion techniques in real time [2][23].

Second, the system must ensure the integrity and immutability of security logs. An unalterable forensic record is essential for accurate incident investigation, compliance, and accountability. This can be achieved through the integration of tamper-resistant technologies, such as blockchain, which provide cryptographic guarantees that logged events cannot be modified or deleted without detection.

Finally, there is a growing need for secure and trustworthy sharing of threat intelligence across organizational boundaries. Current approaches often rely on centralized platforms, which introduce risks of data manipulation and require implicit trust between participants. A decentralized, trustless framework allows multiple organizations to collaboratively share insights—such as attack patterns or model updates—without exposing sensitive data or relying on a single authority.

Together, these requirements define the foundation of a next-generation cybersecurity system: one that proactively detects unknown threats, preserves the integrity of forensic

evidence, and enables secure, decentralized collaboration. Such a system is better equipped to the evolving threat landscape and address the fundamental weaknesses of existing solutions.

1.4 Proposed Solution: The ProSec-Block Framework

This project introduces **ProSec-Block (Proactive Security via Blockchain)**, a unified hybrid cybersecurity framework designed to overcome the key limitations identified in existing research. Unlike prior approaches that address detection, logging, or intelligence sharing in isolation, ProSec-Block integrates all three capabilities into a **single cohesive architecture**, enabling a more efficient and resilient security ecosystem.

At its core, ProSec-Block leverages an **LSTM Autoencoder-based deep learning engine** for behavioral anomaly detection. Instead of relying on predefined attack signatures, the model learns the underlying patterns of normal network behavior over time. By reconstructing expected behavior and measuring deviations, the system can identify anomalies that may indicate malicious activity [23]. This approach is particularly powerful for detecting **zero-day attacks**, as it does not depend on prior knowledge of specific threats but instead focuses on behavioral irregularities that signal potential compromise.[2]

Complementing the AI component, ProSec-Block employs a permissioned blockchain built on Hyperledger Fabric to provide **immutable and verifiable logging** of all detected events. Each alert generated by the AI engine is securely recorded on the blockchain, ensuring that security logs are tamper-proof and auditable. This is critical for forensic analysis, regulatory compliance, and maintaining trust in distributed environments. Additionally, the decentralized nature of the blockchain enables **secure sharing of threat intelligence** across multiple participating organizations without requiring a central authority.

A key strength of ProSec-Block lies in its **proactive detection capability**. By continuously learning and adapting to evolving network behavior, the system can detect emerging threats in real time, significantly reducing response latency [18]. Furthermore, the architecture can be designed to support asynchronous interaction between the AI engine and the blockchain layer, minimizing the performance overhead typically associated with blockchain operations.

1.5 Objectives of the Project

- To design and implement a deep learning-based anomaly detection model (LSTM Autoencoder) capable of identifying zero-day cyber threats in real-time without relying on pre-existing attack signatures [2][23].
- To deploy a permissioned Hyperledger Fabric blockchain network with custom smart contracts (chaincode) for immutable, tamper-proof security event logging and auditing [10][11].
- To develop a decentralized, automated Collaborative Threat Intelligence (CTI) sharing mechanism enabling consortium nodes to exchange verified Indicators of Compromise (IoCs) without a central trusted authority.
- To quantify the reduction in end-to-end incident response time achieved by the proposed integrated AI-Blockchain pipeline compared to conventional architectures [18].
- To evaluate the performance of the system on the publicly available CIC-IDS2019 network intrusion dataset and demonstrate accuracy, precision, recall, and F1-score metrics.

1.6 Scope and Limitations

The scope of this project includes the end-to-end design, development, and evaluation of the ProSec-Block framework within a controlled, simulated environment. The AI-based anomaly detection component is trained and tested using the CIC-IDS2019 dataset, which provides a diverse range of labeled network traffic scenarios representative of modern cyber threats [1]. In parallel, the blockchain layer is implemented and evaluated across a network of five virtual machines, enabling analysis of distributed logging, consensus behavior, and system performance under simulated enterprise conditions [10].

The system is specifically engineered to detect and respond to commonly observed attack categories, including Denial of Service (DoS), Distributed Denial of Service (DDoS), and brute force attacks. These attack types are well-represented in the dataset and provide a practical benchmark for evaluating the effectiveness of the anomaly detection model and the integrity of the logging framework.

However, the project is subject to several important limitations. The most significant is that all evaluations are conducted in a simulated environment rather than a live production network. While simulation allows for controlled experimentation and reproducibility, it cannot fully capture the complexity, variability, and unpredictability of real-world network

traffic and adversarial behavior [6]. As a result, system performance in actual deployment scenarios may differ.

Another key limitation lies in the blockchain layer's consensus latency, which is measured at approximately 2.1 seconds. While acceptable for secure logging and post-incident analysis, this delay may be too high for real-time mitigation of ultra-high-speed attacks, particularly large-scale volumetric DDoS events that can overwhelm systems within milliseconds. This reinforces the current positioning of the blockchain as a trust and audit layer rather than a real-time defensive mechanism [10].

1.7 Relevant Contemporary Issues

The development of ProSec-Block is closely aligned with several pressing challenges in modern cybersecurity, making it both timely and highly relevant. One of the most significant drivers is the **rapid escalation in the global cost of cybercrime**, which is projected to reach trillions of dollars annually. As cyberattacks become more sophisticated and frequent [20], organizations are under increasing pressure to adopt advanced, proactive security mechanisms rather than relying solely on reactive defenses.

At the same time, regulatory frameworks such as General Data Protection Regulation and the Information Technology Act 2000 impose strict obligations on organizations to safeguard user data, ensure privacy, and promptly report security breaches. These regulations elevate the importance of **forensically sound security logging**, where logs must be tamper-proof, verifiable, and auditable [16]. Failure to comply can result in severe legal and financial consequences, further emphasizing the need for robust and trustworthy logging infrastructures.

Another critical factor is the explosive growth of **IoT devices and 5G networks**, which has dramatically expanded the attack surface. Modern networks are no longer centralized; instead, they consist of highly distributed, heterogeneous endpoints ranging from smart devices to industrial systems. Traditional centralized security architectures struggle to scale and adapt to this level of complexity, often becoming bottlenecks or single points of failure [19].

In this context, ProSec-Block's **decentralized architecture** offers a compelling solution. By leveraging blockchain technology, it enables secure, distributed logging and trustless collaboration across multiple nodes, eliminating reliance on a central authority. When combined with AI-driven proactive threat detection, this architecture is inherently better

suited to defend against attacks in large-scale, dynamic environments [5]. It not only enhances resilience and transparency but also aligns with regulatory requirements and the evolving structure of modern network ecosystems, positioning ProSec-Block as a forward-looking solution for next-generation cybersecurity challenges.

1.8 Organization of the Report

The remainder of this report is organized as follows. Chapter 2 presents a comprehensive Literature Survey reviewing the state of the art in AI-based threat detection, blockchain security applications, and their intersection. Chapter 3 describes the Design Flow and Methodology in detail. Chapter 4 presents the Results, Analysis, and Validation including performance metrics and comparative analysis. Chapter 5 presents the Conclusion and outlines directions for Future Work. References, Appendix, and User Manual follow thereafter.

CHAPTER 2 LITERATURE SURVEY

2.1 Introduction to the Literature Review

The domain of cybersecurity is one of the most rapidly evolving fields in all of computer science. To establish a strong theoretical foundation for the proposed ProSec-Block framework, this chapter surveys the existing literature across three interconnected domains: Artificial Intelligence and Machine Learning for cyber threat detection, Blockchain technology applications in security, and hybrid AI-Blockchain systems. The review critically examines the strengths and limitations of each body of work and explicitly links the identified research gaps to the design goals of this project.

2.2 Timeline of Research in Cybersecurity and Intelligent Threat Detection

2.2.1 Era 1 (2000–2010): Rule-Based and Signature-Based Systems

The early 2000s marked a period where **rule-based and signature-based security systems** dominated the cybersecurity landscape. One of the most influential tools of this era was Snort, first released in 1998, which quickly became the de facto standard for open-source intrusion detection. Systems like Snort operated by maintaining extensive databases of known attack signatures—essentially predefined patterns that represent malicious behavior—and scanning network traffic in real time to find matches.

The core strength of this approach lay in its **high accuracy for known threats**. Because signatures were crafted based on previously analyzed attacks, these systems could reliably detect and flag malicious activity that matched existing patterns. This made them particularly effective in enterprise environments where known vulnerabilities and exploits were common. Additionally, their deterministic nature meant fewer false positives compared to more probabilistic detection methods.

However, these systems had significant limitations. They were inherently **reactive rather than proactive**, meaning they could only detect attacks that had already been identified and documented. Any new or unknown threat—commonly referred to as a zero-day attack—would bypass detection entirely until a new signature was developed and deployed. This created a continuous dependency on **manual updates by security experts**, who had to analyze emerging threats and regularly update the signature databases.

Furthermore, as network speeds and data volumes increased, maintaining and efficiently matching against large signature databases became more computationally demanding. This scalability challenge, combined with the inability to detect novel attacks, eventually drove

the shift toward more adaptive and intelligent approaches, such as anomaly-based detection and machine learning models. These newer paradigms aim to identify unusual patterns in behavior rather than relying solely on predefined signatures, addressing many of the shortcomings of early rule-based systems.

2.2.2 Era 2 (2010–2017): Machine Learning for Anomaly Detection

The second era of intrusion detection marked a significant shift from rigid, rule-based systems toward more adaptive and data-driven approaches through the adoption of classical Machine Learning techniques. Researchers began applying supervised learning algorithms such as Decision Trees, Support Vector Machines (SVMs), and Naive Bayes classifiers to analyze network traffic and classify it as either benign or malicious. Unlike earlier signature-based systems, these models learned patterns directly from data, enabling them to generalize beyond explicitly defined rules.

To support experimentation and benchmarking, datasets like KDD Cup 1999 and NSL-KDD became widely adopted as standard evaluation platforms. These datasets provided labeled examples of both normal and attack traffic, allowing researchers to train and compare different models under consistent conditions. The widespread use of these benchmarks helped accelerate progress and establish baseline performance metrics for intrusion detection systems.

The key insight of this era was that Machine Learning could enable **anomaly detection without relying on predefined signatures**. By learning statistical patterns and relationships within the data, ML models could identify unusual or suspicious behavior, including previously unseen attacks. This represented a major advancement over purely reactive systems, introducing a level of adaptability and intelligence into cybersecurity defenses.

2.2.3 Era 3 (2017–Present): Deep Learning and Behavioral Analytics

The current era of cybersecurity is defined by the rapid and widespread adoption of **deep learning techniques**, which have significantly advanced the capabilities of intrusion detection systems. Unlike classical machine learning approaches that relied heavily on handcrafted features, modern models such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, and Transformer-based architectures automatically learn hierarchical representations directly

from raw or minimally processed data. This allows them to capture complex spatial and temporal patterns in network traffic with far greater accuracy and flexibility.

A critical shift in this era is the move away from purely supervised, feature-engineered classification toward **unsupervised and self-supervised behavioral modeling**. These approaches focus on learning the normal behavior of a system and identifying deviations as potential anomalies. As a result, they are far more effective at detecting **zero-day attacks and previously unseen threat patterns**, addressing one of the most significant limitations of earlier systems. Techniques such as autoencoders, generative models, and sequence prediction frameworks enable continuous learning and adaptation, making security systems more resilient in dynamic environments.

At the same time, the emergence of **blockchain technology** has introduced new possibilities for building decentralized and trustworthy security infrastructures. Blockchain provides properties such as immutability, transparency, and distributed consensus, which are particularly valuable for secure logging, audit trails, and collaborative threat intelligence sharing. By eliminating the need for a central authority, blockchain enables multiple organizations to share and verify security data in a trustless manner, enhancing cooperation without compromising data integrity.

2.3 AI and Machine Learning for Threat Detection

2.3.1 Supervised Learning Approaches

Ahmed et al. [1] conducted a comprehensive survey of network anomaly detection techniques, covering supervised algorithms including Random Forest, SVM, and k-Nearest Neighbors. The study concluded that supervised models achieve high accuracy for detecting known attack categories but exhibit a fundamental weakness: they can only identify attacks whose patterns are represented in training data. Kumar and Zhang [6] quantified the false positive and interpretability challenges specific to deep learning models. In production environments, excessive false alarms lead to "alert fatigue" where security operations personnel begin to ignore alerts.

Building on these insights, Kumar and Zhang analyzed the practical deployment challenges of deep learning-based detection systems. Their work emphasizes that even highly accurate models can struggle in production environments due to elevated false positive rates and lack of interpretability. In real-world Security Operations Centers (SOCs), excessive false alerts can overwhelm analysts, leading to a phenomenon known as alert fatigue, where critical warnings may be ignored or overlooked.

Key Points:

- Supervised models (Random Forest, SVM, k-NN) perform well for known attack patterns but fail to generalize to unseen threats.
- Dependence on labeled datasets limits adaptability in dynamic and evolving threat landscapes.
- Deep learning models often suffer from high false positive rates in real-world deployments.
- Lack of model interpretability makes it difficult for analysts to trust and validate alerts.
- Alert fatigue arises when too many false alarms are generated, reducing overall security effectiveness.

2.3.2 Unsupervised and Anomaly-Based Approaches

Khan, Laghari, and Shaikh [2] proposed a deep learning-based approach using autoencoders for network anomaly detection, focusing on unsupervised learning to address the limitations of signature-based systems. In this method, the autoencoder is trained exclusively on benign (normal) network traffic, allowing it to learn a compact representation of typical behavioral patterns. During operation, the model attempts to reconstruct incoming traffic flows; since it has only learned normal patterns, it can reconstruct benign traffic with low error. However, when anomalous or malicious traffic is encountered, the reconstruction error increases significantly. By applying a threshold to this reconstruction error, the system can flag deviations as potential threats—without requiring any labeled attack data. This makes the approach particularly effective for detecting zero-day attacks and previously unseen intrusion patterns.

However, the study also highlights an important limitation of this technique. In real-world networks, legitimate traffic can sometimes exhibit unusual or rare behavior due to sudden workload changes, configuration updates, or atypical user activity. These benign but uncommon patterns may not have been adequately represented during training, leading the model to misclassify them as anomalies. As a result, the system can produce elevated false positive rates, which may reduce its practical effectiveness and increase the burden on security analysts.

Key Points:

- Uses unsupervised learning (autoencoders) trained only on benign traffic.
- Detects anomalies via high reconstruction error.
- Does not require labeled attack data, enabling zero-day detection.
- Effective for identifying previously unseen threats.
- Limitation: high false positives when normal traffic shows unusual but legitimate variations.

2.3.3 Temporal and Sequential Models

Singh and Patel [3] argued that modern network attacks are not isolated events but multi-stage processes that unfold over time, with each phase building on the previous one. Traditional signature-based systems typically detect attacks only at their final stages—such as data exfiltration or system compromise—by which point significant damage may have already occurred. This delayed detection highlights a critical gap in conventional approaches, which fail to account for the temporal progression of attacks.

To address this, they proposed the use of Long Short-Term Memory (LSTM) networks, which are specifically designed to capture long-range temporal dependencies in sequential data. By analyzing sequences of network traffic flows over time, LSTMs can identify subtle patterns that emerge during earlier stages of an attack, such as reconnaissance (e.g., scanning activities) and lateral movement across hosts. This enables earlier detection, potentially stopping an attack before it reaches its most damaging phase.

Their work directly influenced the architectural design of ProSec-Block, where LSTM layers are used as the core temporal modeling component in the anomaly detection engine. By incorporating temporal awareness into the system, ProSec-Block moves beyond static or point-in-time analysis and instead evaluates how network behavior evolves, significantly enhancing its ability to detect sophisticated, multi-stage cyber threats.

Key Points:

- Network attacks are multi-stage and time-dependent, not isolated events.
- Signature-based systems detect attacks too late, often after damage is done.
- LSTM networks capture long-range temporal dependencies in traffic data.
- Enables detection during early phases like reconnaissance and lateral movement.

- Inspired the use of LSTM layers in ProSec-Block’s detection engine.

2.3.4 Transformer-Based Architectures

Chen, Zhang, and Wang [5] extended the capabilities of the Transformer architecture into the domain of network intrusion detection by reinterpreting network traffic in a novel way. Instead of treating packets or flows as isolated events, they modeled each network flow as a “token,” similar to words in a sentence in natural language processing. This allowed them to leverage multi-head self-attention mechanisms to learn complex dependencies and contextual relationships across sequences of network activity. By doing so, their approach was able to capture long-range interactions and subtle behavioral patterns that are often missed by traditional sequence models. Experimental results demonstrated that their Transformer-based model significantly outperformed conventional LSTM-based approaches on widely used intrusion detection benchmarks, particularly in detecting sophisticated and previously unseen attack patterns.

However, despite its strong performance, the model introduced several practical limitations. Transformer architectures are inherently data-hungry and require large-scale labeled datasets to generalize effectively. Additionally, the computational overhead associated with multi-head attention, high-dimensional embeddings, and deep stacking of layers results in increased training time and memory consumption. This makes deployment challenging in real-world scenarios where computational resources, latency requirements, or energy efficiency are critical concerns—such as edge devices, IoT environments, or real-time monitoring systems.

Considering these constraints, the LSTM Autoencoder was chosen as the primary architecture for this project. LSTM-based models are well-suited for sequential data and can effectively learn temporal patterns with comparatively lower computational requirements. The autoencoder structure further enables unsupervised learning, allowing the model to detect anomalies by reconstructing normal behavior and identifying deviations without relying heavily on labeled data. This makes the LSTM Autoencoder a more practical and efficient choice for environments with limited resources, while still maintaining strong performance in anomaly detection tasks.

2.4 Blockchain Technology in Security Applications

2.4.1 Foundational Blockchain Concepts

The foundational concept of a decentralized, peer-to-peer electronic ledger was formally introduced by Satoshi Nakamoto in the context of the Bitcoin system. In this model, data is organized into a chain of blocks, where each block contains a timestamp, a collection of transactions, and a cryptographic hash of the previous block. This **hash-linking mechanism** ensures that all blocks are interdependent—any modification to the contents of a historical block changes its hash, which in turn breaks the linkage with all subsequent blocks.

As a result, even a minor attempt to alter past data becomes immediately detectable, since the inconsistency propagates throughout the chain. This property forms the basis of blockchain's **tamper-evident and immutable design**, where trust is established not through a central authority but through cryptographic verification and distributed consensus among network participants.

While the original application of this architecture was focused on enabling secure and transparent financial transactions, its underlying principles have far-reaching implications beyond cryptocurrency. In particular, the ability to maintain **immutable, time-stamped, and verifiable records** makes blockchain highly suitable for applications such as security logging, audit trails, and digital forensics. In these contexts, ensuring that logs cannot be altered or deleted without detection is critical for maintaining accountability, supporting investigations, and complying with regulatory requirements. Thus, the blockchain data structure provides a robust foundation for building trustworthy systems in domains where data integrity is paramount.

2.4.2 Immutable Security Logging and Permissioned Blockchains

Dorri, Kanhere, and Jurdak [8] explored the application of blockchain technology in securing IoT device logs, addressing one of the key challenges in distributed environments—ensuring the integrity and trustworthiness of generated data. In their work, they proposed the use of hash-chaining mechanisms, where each log entry is cryptographically linked to the previous one. This creates an immutable sequence of records such that any attempt to alter or tamper with a log entry would break the chain and become immediately detectable. By integrating this approach with blockchain, they demonstrated how decentralized ledger systems can provide strong guarantees of data integrity, transparency, and resistance to unauthorized modifications. Their findings highlight the suitability of blockchain as a secure logging infrastructure for IoT ecosystems, where devices are often resource-constrained and vulnerable to attacks.

Building on the importance of selecting an efficient blockchain platform, Ali and Park [10] conducted a detailed comparative analysis of two widely used frameworks: Ethereum and Hyperledger Fabric, specifically in the context of security logging applications. Their evaluation focused on key performance metrics such as throughput and latency, which are critical for real-time or near real-time logging systems. The results revealed that Hyperledger Fabric, when configured with the Raft consensus algorithm, significantly outperforms Ethereum in terms of scalability and responsiveness. Hyperledger Fabric achieved transaction throughput in the range of 400–600 transactions per second (TPS) with relatively low latency of 1–3 seconds. In contrast, Ethereum demonstrated a much lower throughput of 15–30 TPS and higher latency ranging from 10–60 seconds, primarily due to its proof-of-work-based consensus mechanism and public network overhead.

2.4.3 Decentralized Threat Intelligence Sharing

Sharma, Singh, and Lee [9] introduced a blockchain-based platform for **Cyber Threat Intelligence (CTI) sharing**, aiming to solve one of the major challenges in collaborative cybersecurity—establishing trust among organizations without relying on a centralized authority. In their system, participating entities contribute and access Indicators of Compromise (IoCs), such as malicious IP addresses, domain names, or file hashes, through smart contracts deployed on a blockchain network. By leveraging blockchain’s decentralized and immutable nature, the platform ensures that shared intelligence is transparent, tamper-resistant, and verifiable by all participants, thereby enabling **trustless peer-to-peer information exchange**.

This approach is particularly valuable in multi-organization environments where concerns about data integrity, ownership, and trust often hinder effective collaboration. The use of smart contracts automates access control, submission, and retrieval of IoCs, reducing dependency on centralized intermediaries and lowering the risk of single points of failure. As a result, the system enhances the reliability and accountability of CTI sharing, which is critical for detecting and mitigating large-scale or distributed cyber threats.

However, despite its innovative design, the system has a notable limitation: it depends on **manual validation of IoCs by human analysts** before they are accepted and shared across the network. While this step helps ensure the accuracy and relevance of threat intelligence, it introduces a significant bottleneck. In cybersecurity, where threats evolve rapidly and response time is crucial, delays in validating and propagating IoCs can reduce their effectiveness. Time-sensitive indicators may become obsolete or less impactful by the time they are verified and distributed.

This limitation highlights an important gap that systems like ProSec-Block aim to address. By incorporating **automated validation mechanisms**, potentially driven by AI models, the validation process can be significantly accelerated while maintaining reliability. Automating this step not only reduces human workload but also enables near real-time sharing of threat intelligence, improving the overall responsiveness and effectiveness of collaborative defense strategies.

2.5 AI-Blockchain Hybrid Systems for Cybersecurity

2.5.1 Basic AI-Blockchain Integration

Lee, Kim, and Yoo [11] proposed one of the earliest attempts to **tightly integrate artificial intelligence with blockchain** in a cybersecurity setting, laying important groundwork for subsequent hybrid systems. In their architecture, an AI-based Intrusion Detection System (IDS) is responsible for identifying suspicious or malicious activities in network traffic. Once a threat is detected, instead of storing full logs directly on-chain (which would be inefficient), the system computes a **cryptographic hash** of the detection record and stores this hash on the blockchain. This design choice ensures data integrity while minimizing storage overhead, as the actual detailed logs can remain off-chain.

The key advantage of this approach lies in its ability to provide **tamper-proof forensic evidence**. Even if an attacker later compromises the local system and attempts to alter or delete the original logs, the corresponding hash stored on the blockchain remains immutable. Any discrepancy between the stored hash and the modified log can be easily detected, thereby preserving the trustworthiness of the evidence. This is particularly valuable in incident investigation, compliance auditing, and legal contexts where the authenticity of security logs is critical.

However, their experimental evaluation also revealed an important limitation: the **latency overhead introduced by the blockchain logging step**. Because each detection event requires interaction with the blockchain network—often involving transaction validation and consensus—the process can introduce delays that are non-trivial, especially in high-frequency attack scenarios. When logging is treated as a synchronous operation, it can slow down the overall detection-response pipeline, reducing the system’s effectiveness in real-time environments.

2.5.2 Smart Contract-Based Automated Response

Wang and Liu [12] advanced the integration of artificial intelligence and blockchain by moving beyond passive detection and secure logging toward **automated response**

systems. In their approach, AI models are not only responsible for identifying potential threats but also for assigning a confidence score to each detection. These outputs are then directly integrated with blockchain-based smart contracts (or chaincode), which are programmed to automatically trigger predefined remediation actions when the confidence score exceeds a specified threshold. For example, if suspicious activity is detected with high confidence, the system can immediately initiate countermeasures such as blocking an IP address, isolating a compromised node, or updating firewall rules. This tight coupling between detection and response significantly reduces reaction time and minimizes human intervention, making the system more adaptive and responsive to real-time threats.

However, the authors also emphasized a critical risk associated with this level of automation—the **security of the smart contracts themselves**. Since smart contracts operate autonomously and enforce actions without manual oversight, any vulnerability in their logic can have severe consequences. If an attacker manages to exploit flaws in the chaincode, they could manipulate the system into triggering false remediation actions. For instance, malicious actors could craft inputs that deceive the AI model into producing high-confidence false positives, which in turn would activate the smart contract to block legitimate users or services. Such scenarios could effectively result in a **self-inflicted denial-of-service (DoS)**, where the system disrupts its own normal operations.

Therefore, while automated AI-blockchain integration enhances efficiency and responsiveness, Wang and Liu highlight the importance of implementing rigorous smart contract validation, secure coding practices, and robust testing mechanisms. Safeguards such as multi-factor verification, anomaly thresholds, human-in-the-loop controls, or fallback mechanisms can help mitigate these risks. Their work underscores that in security-critical systems, automation must be carefully balanced with reliability and resilience to avoid unintended consequences.

network.

2.5.3 Federated Learning with Blockchain Aggregation

Garcia, Lopez, and Fernandez [13] investigated the integration of **federated learning with blockchain technology** to address privacy concerns in collaborative intrusion detection systems. In their approach, multiple organizations independently train local AI models using their own network traffic data, ensuring that sensitive information never leaves their internal environments. Instead of sharing raw data, only model updates—typically in the form of gradients or weights—are transmitted. These updates are then recorded and

managed through a blockchain network, which ensures transparency, immutability, and trust among participating entities that may not fully trust each other.

A key component of their framework is the use of a **global model aggregation smart contract**, which is responsible for combining the locally trained model updates into a single, improved global model. This decentralized aggregation mechanism eliminates the need for a central authority while still enabling collaborative learning. As a result, organizations benefit from collective intelligence, improving detection accuracy across diverse attack patterns without compromising data privacy. This approach is particularly valuable in cybersecurity domains, where sharing raw network traffic data is often restricted due to legal, regulatory, or competitive concerns.

However, the authors identified a major limitation in their system—**significant communication overhead**. Exchanging model gradients via blockchain transactions introduces latency and increases network load, especially when dealing with large models or frequent updates. Since blockchain networks are not inherently optimized for high-volume data transfer, this can lead to scalability issues and reduced efficiency. The overhead becomes even more pronounced as the number of participating organizations grows, making it challenging to maintain real-time or near real-time performance.

2.6 Comparative Analysis of Related Work

Table 2.1: Comprehensive Comparative Analysis of Related Literature

Ref.	Authors & Year	Technology Used	Key Contribution	Major Limitation
[1]	Ahmed et al., 2023	Supervised ML (RF, SVM)	Survey of anomaly detection; High accuracy for known attacks	Fails on zero-day attacks
[2]	Khan et al., 2024	Deep Autoencoder (Unsupervised)	Detects novel threats via reconstruction error	High false positive rates
[3]	Singh & Patel, 2025	LSTM (Temporal ML)	Early detection of multi-stage attacks	Computationally intensive
[5]	Chen et al., 2025	Transformer-based IDS	Outperforms RNNs for network intrusion	Requires large training data
[6]	Kumar & Zhang, 2024	DL Analysis	Quantifies false positive and interpretability problems	No proposed solution
[8]	Dorri et al., 2023	Blockchain for IoT Logging	Ensures log immutability	Scalability concerns on-chain
[9]	Sharma et al., 2024	Blockchain CTI Platform	Decentralized incentivized IoC sharing	Manual IoC validation bottleneck

[10]	Ali & Park, 2025	Hyperledger Fabric vs. Ethereum	Permissioned BC offers high throughput	Requires trusted membership
[11]	Lee et al., 2024	AI + Blockchain	Tamper-proof AI evidence logging	High end-to-end latency
[12]	Wang & Liu, 2025	AI + Smart Contracts	Automated remediation triggered by AI	Smart contract security risks
[13]	Garcia et al., 2025	Federated Learning + BC	Privacy-preserving collaborative model training	High communication overhead

2.7 Research Gap Analysis

A careful synthesis of the surveyed literature highlights a clear and persistent research gap: the absence of a **fully integrated, end-to-end AI–Blockchain framework** that simultaneously addresses proactive threat detection, tamper-proof logging, and collaborative intelligence within a single cohesive system [17][21]. While significant progress has been made in each of these domains individually, existing approaches largely remain fragmented, limiting their effectiveness in real-world cybersecurity deployments.

On one hand, AI-based intrusion detection research has primarily focused on improving model accuracy, precision, and recall. Techniques such as deep learning and sequence modeling have demonstrated strong capabilities in identifying complex and previously unseen attack patterns [2][3]. However, these efforts often overlook how detection outputs are securely stored, audited, or shared across systems. The lack of integration with robust logging mechanisms means that even accurate detections may not be reliably preserved for forensic analysis or regulatory compliance.

On the other hand, blockchain research in the context of cybersecurity has concentrated on enhancing data integrity, transparency, and decentralized trust. Solutions emphasize high transaction throughput, efficient consensus mechanisms, and secure logging architectures [8][17]. Yet, these systems typically treat logged data as static records, without leveraging it to continuously improve or adapt AI-driven detection models. As a result, the potential feedback loop between secure data storage and intelligent threat detection remains underexplored[11].

Although a few hybrid AI-blockchain systems have been proposed, they often suffer from **prohibitive end-to-end latency**. This is largely due to the design choice of treating blockchain logging as a synchronous, blocking operation within the detection pipeline. Such an approach introduces delays that are incompatible with real-time or near real-time

security requirements, undermining the benefits of rapid AI-based detection [10][6]. Consequently, these systems struggle to scale effectively in dynamic and high-throughput network environments.

Furthermore, a critical limitation across the literature is the lack of **comprehensive performance evaluation at the system level**. Specifically, no existing study has rigorously measured and reported the **end-to-end incident response time improvements** achieved by integrating proactive AI detection with blockchain-based logging and coordination [18]. Without such evaluation, it is difficult to quantify the practical benefits of these hybrid architectures or justify their deployment in operational settings.

This gap underscores the need for a novel framework that not only integrates AI and blockchain technologies but also optimizes their interaction to minimize latency, enable seamless data flow, and support collaborative intelligence. Such a system should decouple detection from logging through asynchronous mechanisms, enable feedback-driven model updates, and provide measurable improvements in incident response time—thereby addressing the limitations of current approaches and advancing the state of the art in cybersecurity systems design.

2.8 Linking Literature to the Problem Statement

The literature review provides a strong foundation for the three-part problem statement presented in Chapter 1, clearly demonstrating that current cybersecurity solutions are fragmented and insufficient when applied to real-world environments [1][17]. Existing research tends to address individual challenges—such as threat detection, secure logging, or intelligence sharing—in isolation, rather than as components of a unified system. This disconnect results in gaps in performance, security, and scalability. By critically analyzing prior work, this project identifies these limitations and integrates their strengths into a single architecture, ProSec-Block, which holistically addresses proactive detection, tamper-proof logging, and collaborative intelligence.

Key Supporting Points:

- **Zero-Day Threat Detection Limitation:**

Traditional rule-based and signature-based systems are ineffective against unknown or evolving threats. This limitation necessitates the use of behavioral anomaly detection techniques as explored in [2] and [3]. ProSec-Block adopts this approach through an LSTM Autoencoder, enabling the system to detect anomalies based on

deviations from learned normal behavior rather than relying on predefined signatures.

- **Vulnerability of Centralized Logging Systems:**

Centralized security logs are prone to tampering, deletion, and unauthorized modification, especially after a system compromise. Research such as [8] and [10] highlights the use of blockchain to ensure data immutability and integrity. Based on this, ProSec-Block incorporates a blockchain logging layer using Hyperledger Fabric, ensuring that all threat records are securely stored and verifiable.

- **Limitations of Centralized CTI Sharing:**

Existing Cyber Threat Intelligence platforms rely on centralized authorities, leading to trust issues and single points of failure. Studies like [9] and [12] propose decentralized sharing using blockchain and smart contracts. ProSec-Block extends this idea by enabling trustless, peer-to-peer IoC sharing across organizations.

- **Lack of Automation in Existing Systems:**

Many existing CTI sharing systems depend on manual validation of threat intelligence, creating delays in response [24]. ProSec-Block addresses this by introducing automated validation and integration of IoCs into the AI engine, reducing response time and improving efficiency.

- **Absence of Feedback Loop in Hybrid Systems:**

Most prior hybrid AI-blockchain systems lack a mechanism for continuous learning. ProSec-Block fills this gap by implementing a feedback loop where shared IoCs are used to update blacklists and fine-tune the LSTM Autoencoder, ensuring continuous adaptation to new threats [13][21].

- **Need for an Integrated Architecture:**

The literature reveals that no existing system fully integrates detection, logging, and intelligence sharing [17][21]. ProSec-Block bridges this gap by combining all three components into a cohesive, low-latency framework, making it more practical for real-world deployment.

CHAPTER 3 DESIGN FLOW AND METHODOLOGY

3.1 System Overview

This chapter provides a detailed and comprehensive description of the ProSec-Block framework's design, architecture, algorithms, and implementation. The system is conceived as a modular pipeline comprising four major layers that communicate through well-defined interfaces: the Data Acquisition and Preprocessing Layer, the AI-based Proactive Detection Layer, the Blockchain Integration and Logging Layer, and the Collaborative Threat Intelligence and Feedback Layer. The design philosophy underpinning ProSec-Block is the principle of "defence in depth through intelligence [20]": rather than relying on a single point of control, the system distributes security intelligence across multiple nodes.

3.2 Design Constraints and Considerations

3.2.1 Technical Constraints

The system is required to meet strict performance and reliability standards to ensure it is practically useful in real-world enterprise security environments. One of the most critical requirements is detection latency. The system must be capable of identifying and responding to threats with minimal delay, ideally generating a logged alert within 5 seconds from the moment the first anomalous packet is observed. This low latency is essential because even a small delay in detection can allow malicious activities—such as data exfiltration, lateral movement, or privilege escalation—to progress significantly, increasing potential damage [18]. Achieving this level of responsiveness requires optimized data pipelines, efficient feature extraction, and fast inference from the AI model.

In addition to detection speed, the blockchain layer must be designed to handle high-throughput conditions without becoming a bottleneck. Enterprise networks can generate a substantial number of security-related events, particularly during peak traffic periods or under active attack scenarios. The system is expected to sustain a throughput of approximately 400–600 events per second. This means the blockchain infrastructure must efficiently validate, process, and record each event in a tamper-proof manner while maintaining consistency and low latency [10]. Any backlog in processing could result in delayed logging, reduced visibility, and potential loss of critical forensic data. Therefore, scalability mechanisms such as parallel processing, optimized consensus protocols, or layer-2 solutions may be necessary to maintain performance under load.

Equally important is the accuracy of the AI model used for anomaly detection. The system must maintain a False Positive Rate (FPR) below 2%, which is crucial for operational

effectiveness. A high rate of false positives can overwhelm security analysts with unnecessary alerts, leading to alert fatigue and increasing the likelihood that genuine threats may be overlooked or ignored [6]. Maintaining a low FPR requires careful model training, high-quality labeled datasets, and continuous tuning based on real-world feedback. At the same time, the model must balance this with a high detection rate to ensure that actual threats are not missed.

Overall, the system must strike a balance between speed, scalability, and accuracy. Low detection latency ensures rapid response, high blockchain throughput guarantees reliable and timely logging of events, and a low false positive rate ensures that security teams can trust and effectively act on the alerts generated. Together, these requirements define a robust, efficient, and operationally viable security monitoring solution suitable for modern enterprise environments.

3.2.2 Economic and Practical Constraints

The proposed framework is designed with practicality and accessibility in mind, ensuring that it can be deployed on commodity hardware typically available in enterprise data centres. This means organizations do not need to invest in highly specialized or expensive infrastructure to adopt the system. By leveraging standard servers with moderate CPU, memory, and storage capabilities, the framework becomes easier to integrate into existing IT environments without significant capital expenditure. This approach also simplifies scalability, as additional nodes or resources can be added incrementally using readily available hardware.

A key design choice is the reliance on open-source technologies such as TensorFlow and Keras for the AI components, along with Hyperledger Fabric for the blockchain layer. The use of these widely adopted tools eliminates licensing costs, which can otherwise be a major barrier to adoption, especially for small and medium-sized enterprises. Additionally, open-source ecosystems provide strong community support, regular updates, and flexibility for customization, allowing organizations to tailor the framework according to their specific security requirements.

At the same time, the framework must carefully balance its operational overhead with the value it delivers. Running an AI inference engine continuously, along with maintaining a distributed blockchain network, inevitably consumes computational resources such as CPU cycles, memory, and network bandwidth. Therefore, the system must be optimized to ensure that these resource requirements remain reasonable [6] and do not strain existing

infrastructure. Techniques such as lightweight model architectures, efficient batching of inference tasks, and optimized blockchain consensus mechanisms can help reduce overhead.

Ultimately, the computational and operational costs associated with the framework must be justified by the security benefits it provides. These benefits include faster threat detection, tamper-proof logging of security events, improved auditability, and enhanced trust in the system's outputs. When the reduction in risk, improved incident response time, and stronger compliance capabilities are considered, the overall trade-off becomes favorable, making the framework both economically and operationally viable for enterprise deployment.

3.2.3 Ethical and Legal Constraints

The system is designed with privacy and regulatory compliance as a core requirement, particularly because network traffic can inherently contain personally identifiable information (PII). To align with data protection frameworks such as the General Data Protection Regulation and the Information Technology Act, 2000, the architecture deliberately avoids intrusive inspection of user data [16].

Instead of performing deep packet inspection (DPI), which involves analyzing the actual payload content of network communications, the system operates exclusively on flow-level features. These features are statistical aggregates derived from packet streams, such as flow duration, packet counts, byte volumes, and timing characteristics. By focusing on behavioral metadata rather than content, the system ensures that sensitive information—such as messages, credentials, or personal data contained within packets—is never accessed, processed, or stored.

This design choice provides a strong privacy-preserving advantage. Since only abstracted patterns of communication are analyzed, the risk of exposing or mishandling PII is significantly reduced. At the same time, these flow-level characteristics are still rich enough to enable effective anomaly detection, allowing the AI model to identify suspicious patterns such as unusual traffic spikes, irregular timing behavior, or abnormal connection patterns.

Ultimately, this approach strikes a careful balance between security and privacy. It enables robust threat detection capabilities while ensuring compliance with legal and ethical standards, reducing the risk of regulatory violations and building greater trust in the system's operation.

3.3 Alternative Design Approaches Considered

3.3.1 Alternative Design A: Centralized AI with Public Blockchain

The first alternative design explored in the system architecture was a model in which a centralized AI server is responsible for analyzing all incoming network traffic and subsequently logging detected security events onto a public blockchain such as Ethereum. At first glance, this approach offers a high degree of simplicity in terms of system design, as all intelligence and decision-making are concentrated in a single location. Additionally, by leveraging a public blockchain, it appears to maximize decentralization and transparency, ensuring that recorded events are immutable and verifiable by any participant [7].

However, despite these apparent advantages, the design was ultimately rejected due to several critical limitations that undermine its suitability for enterprise environments. The most significant concern is the reliance on a centralized AI server. This creates a single point of failure within the system, meaning that if the server is compromised, overloaded, or taken offline, the entire detection and logging pipeline collapses [20]. From a security perspective, this also makes the server a highly attractive target for attackers, as successfully breaching it could disable monitoring capabilities or allow manipulation of detection outcomes.

Another major drawback lies in the performance limitations of public blockchain networks like Ethereum. With a transaction throughput typically ranging between 15 to 30 transactions per second, such platforms are not capable of handling the high volume of security events generated in enterprise networks, which can reach hundreds of events per second [10]. This mismatch would inevitably lead to transaction backlogs, increased latency, and potentially dropped or delayed logs, all of which are unacceptable in a real-time security monitoring system.

3.3.2 Alternative Design B: Distributed AI Agents with Centralized Database

The second alternative design considered was a fully distributed swarm of lightweight AI monitoring agents deployed across the network, each responsible for local traffic analysis and anomaly detection. These agents would report their findings to a centralized SQL database for log aggregation and threat intelligence sharing, making the overall architecture operationally similar to traditional Security Information and Event Management (SIEM) platforms such as Splunk. At a surface level, this design offers advantages in terms of

scalability and distributed data collection, as multiple agents can independently monitor different segments of the network, reducing the load on any single node.

However, despite these benefits, the architecture was ultimately rejected due to a fundamental weakness: it does not adequately solve the problem of log integrity. By relying on a centralized SQL database as the core repository for security logs, the system introduces a critical vulnerability. If an attacker gains sufficient privileges—either through credential compromise, insider threats, or exploitation of system vulnerabilities—they could alter, delete, or inject log entries. This undermines the trustworthiness of the entire monitoring system, as logs are often used for forensic analysis, compliance audits, and incident response [20].

In addition to the integrity issue, the centralized threat intelligence repository presents another significant risk. Because all agents depend on this central source for shared intelligence, it becomes a high-value target for adversaries. An attacker who successfully compromises this repository could perform intelligence poisoning—injecting false or misleading data that propagates across all monitoring agents [9]. This could lead to incorrect threat assessments, missed detections, or even deliberate misdirection of security teams.

Thus, while the distributed agent model improves data collection coverage, its reliance on centralized storage and intelligence sharing creates structural weaknesses. The risks of log tampering and intelligence poisoning, combined with the presence of a single critical attack surface, make this design unsuitable for environments where trust, integrity, and resilience are paramount.

3.3.3 Selected Design: ProSec-Block (Distributed AI + Permissioned Blockchain)
The final selected design addresses the shortcomings of both alternatives. Distributed AI agents eliminate the single point of failure. The permissioned Hyperledger Fabric blockchain provides high throughput and confidentiality while offering tamper-evidence and decentralization [10][11]. The comparison table below summarizes the trade-off analysis that led to this selection.

Table 3.1: Comparative Evaluation of Alternative Designs

Criterion	Design A: Centralized AI + Public BC	Design B: Distributed AI + Central DB	Selected: ProSec-Block
Zero-Day Detection	Yes (AI)	Yes (AI)	Yes (LSTM Autoencoder)
Log Immutability	Yes (Public BC)	No (Central DB)	Yes (Hyperledger Fabric)

Scalable Throughput	No (~30 TPS)	Yes (DB scales)	Yes (~450 TPS)
Data Confidentiality	Low (Public BC)	Medium (Centralized)	High (Permissioned)
No Single Point of Failure	Partial (AI SPOF)	No (DB and AI SPOF)	Yes (Fully distributed)
CTI Sharing Trust	Medium	Low	High (Smart Contracts)

3.4 System Architecture: ProSec-Block

The ProSec-Block architecture consists of four principal layers, each designed as an independent module communicating with adjacent layers through defined APIs, enabling each component to be upgraded or replaced independently without disrupting the rest of the system.

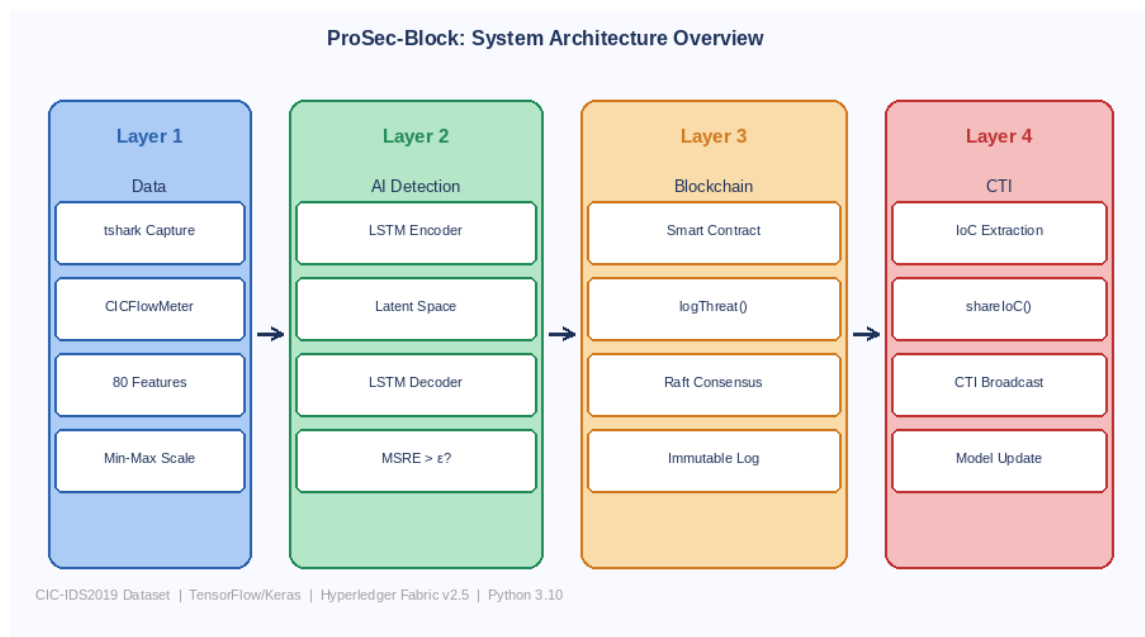


Figure 3.1: Detailed System Architecture of the ProSec-Block Framework

3.5 Layer 1: Data Acquisition and Preprocessing

3.5.1 Network Traffic Capture and Feature Extraction

Real-time network traffic capture in the proposed framework is carried out using tshark, a command-line utility derived from Wireshark that enables efficient and continuous monitoring of packets across a specified network interface. It is configured to capture all incoming and outgoing packets in real time, ensuring that no potentially relevant traffic is missed. This low-level packet acquisition forms the foundational data layer of the system, providing a complete and unfiltered view of network activity as it occurs.

Once the raw packet stream is collected, it is passed to CICFlowMeter for structured processing. This tool transforms the continuous packet stream into bidirectional network flows, where each flow represents a sequence of packets exchanged between a source and a destination. By grouping packets into flows, the system moves from raw, unstructured data to a more meaningful representation of network communication patterns, which is better suited for analysis by machine learning models [18].

During this transformation, CICFlowMeter extracts a comprehensive set of 80 statistical features for each flow. These features are carefully designed to capture the behavioral characteristics of network interactions without inspecting the actual payload content, thereby preserving privacy and reducing computational complexity [19]. Key features include flow duration, total forward and backward packet counts, total byte volumes, and detailed inter-arrival time statistics such as minimum, maximum, mean, and standard deviation. In addition, protocol-specific indicators like TCP flag counts (e.g., SYN, ACK, FIN flags) and window size statistics are also recorded, providing insights into connection behavior and potential anomalies.

3.5.2 Preprocessing Pipeline

The 80 extracted features undergo a two-stage preprocessing pipeline. First, categorical features (protocol type: TCP, UDP, ICMP) are one-hot encoded into binary vectors. Second, all continuous numerical features are normalized to the range [0, 1] using Min-Max scaling:

$$\text{Max scaling: } x_{\text{normalized}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

The normalization statistics are computed from the training set and saved for consistent application during inference. The preprocessed feature vectors are grouped into sliding windows of 100 consecutive flows, creating temporal sequences for input to the LSTM-based model [23].

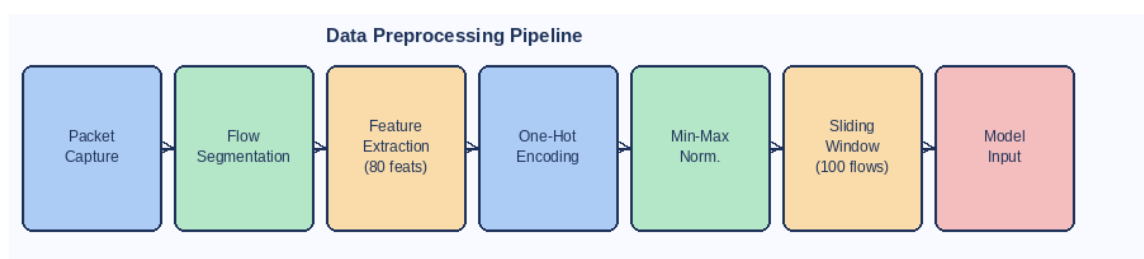


Figure 3.2: Data Preprocessing Pipeline Flowchart

3.6 Layer 2: AI-Based Proactive Detection Engine

3.6.1 Model Architecture: LSTM Autoencoder

The core of the detection engine is an LSTM Autoencoder, chosen over simple feedforward autoencoders because of its ability to model temporal dependencies in sequential data — critical for detecting the multi-stage, time-ordered patterns characteristic of sophisticated network attacks [3][23]. The architecture consists of an encoder that compresses the input sequence into a latent representation and a decoder that attempts to reconstruct the original sequence from this compressed form.

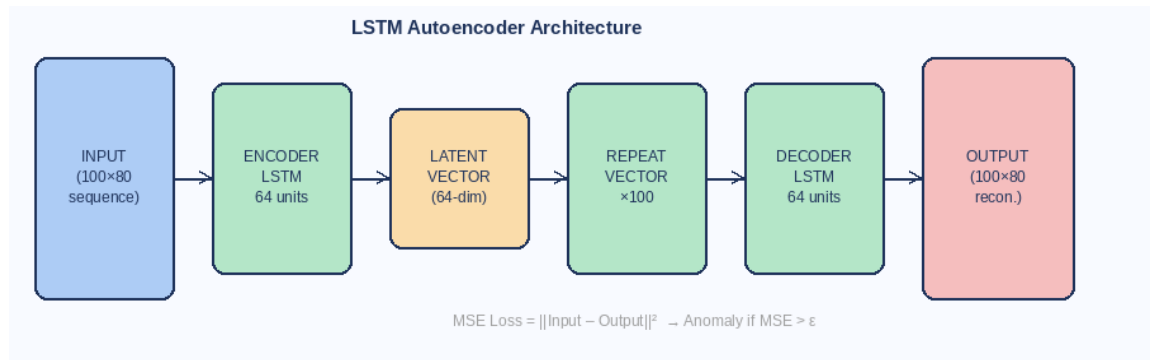


Figure 3.3: LSTM Autoencoder Architecture — Encoder, Latent Vector, Decoder, and Reconstruction

Table 3.2: LSTM Autoencoder Architecture Specification

Layer	Type	Units/Size	Activation	Notes
1	LSTM (Encoder)	64 units	tanh	Returns sequence=False; captures temporal context
2	RepeatVector	100	N/A	Repeats the latent vector for decoder input
3	LSTM (Decoder)	64 units	tanh	Returns sequence=True
4	TimeDistributed Dense	80	Linear	Reconstructs each timestep's 80 features

3.6.2 Training Procedure

The model is trained exclusively on the benign traffic portion of the CIC-IDS2019 dataset (80% of benign flows for training, 20% for validation). The loss function is Mean Squared Error (MSE) between the original input sequence and the reconstructed output. The Adam optimizer is used with a learning rate of 0.001. Training is conducted for 50 epochs with a batch size of 256, using early stopping (patience = 5 epochs) to prevent overfitting. The training set contains no attack traffic — this fundamental design choice enables zero-day detection [3][23].

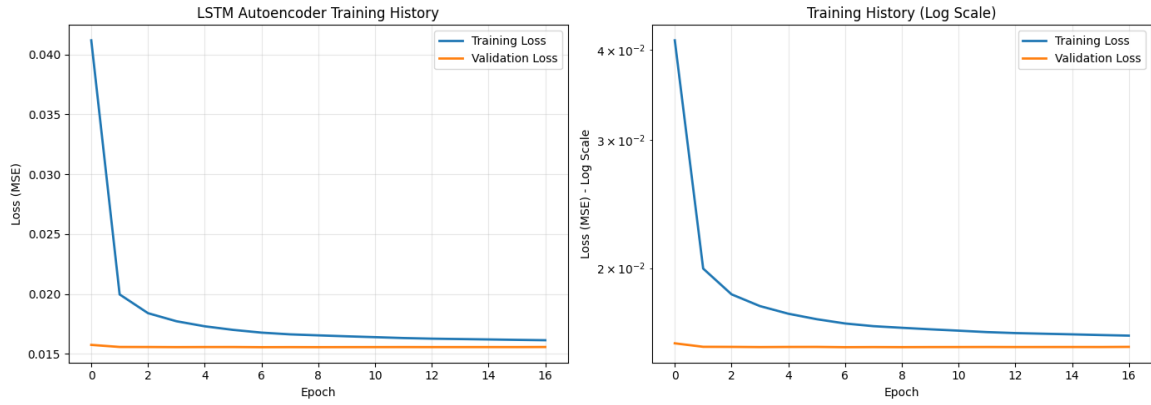


Figure 3.4: Training and Validation Loss Curves for the LSTM Autoencoder (50 Epochs)

3.6.3 Anomaly Detection Mechanism

After the training phase, the LSTM Autoencoder is deployed for inference by processing incoming network data in fixed-length sequences of 100 flows. For each sequence, the model attempts to reconstruct the input and computes the **Mean Squared Reconstruction Error (MSRE)**, which quantifies how closely the reconstructed output matches the original input. Since the model has been trained primarily on benign traffic, it is expected to reconstruct normal patterns with low error, while anomalous or malicious patterns will result in significantly higher reconstruction errors [2].

To systematically distinguish between normal and anomalous behavior, an anomaly threshold ε is defined using the validation dataset consisting of benign traffic. Specifically, the threshold is set at the **95th percentile of the MSRE distribution**, meaning that 95% of reconstruction errors for normal data fall below this value. This statistically grounded approach ensures that only the top 5% of unusually high errors—relative to known benign behavior—are considered suspicious.

$$\varepsilon = \text{Percentile}_{95}(\text{MSRE}_{\text{benign}})$$

During inference, any sequence whose MSRE exceeds this threshold ($\text{MSRE} > \varepsilon$) is flagged as a potential anomaly. This method provides a **balanced trade-off between sensitivity and specificity**. By choosing the 95th percentile, the system maintains a high detection rate (sensitivity) for anomalous events while controlling the false positive rate by not over-flagging normal variations in traffic. This principled thresholding strategy ensures that the anomaly detection mechanism remains both effective and practical for real-world deployment, where excessive false alarms can be as problematic as missed detections [23].

(false positive rate).

3.7 Layer 3: Blockchain Integration for Immutable Logging

3.7.1 Platform Selection and Network Topology

Hyperledger Fabric v2.5 was selected as the blockchain infrastructure due to its Raft consensus mechanism offering 400–600 TPS with finality times of 1–3 seconds — far superior to public blockchains for enterprise security logging. The ProSec-Block Hyperledger Fabric network consists of five Peer Nodes (security monitoring agents), five Orderer Nodes running Raft consensus, and a Certificate Authority (CA) for managing node identity and access control. The five-node Raft ordering service provides crash fault tolerance: the network continues to operate correctly even if up to two orderer nodes fail simultaneously.

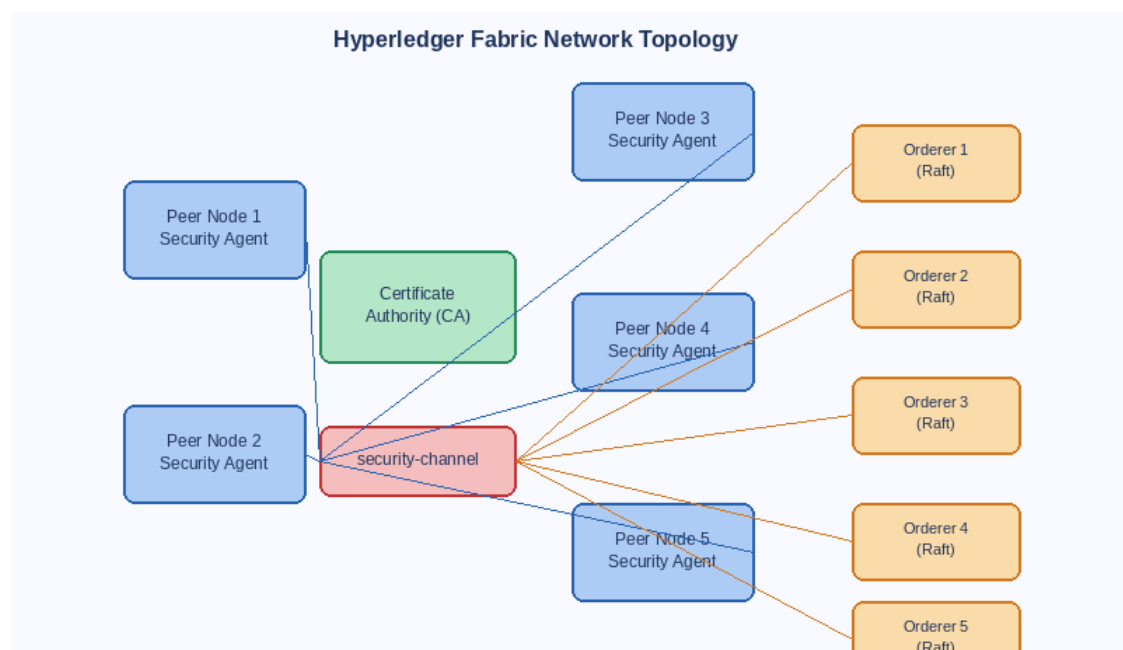


Figure 3.5: Hyperledger Fabric Network Topology for ProSec-Block

3.7.2 Smart Contract (Chaincode) Design

Two chaincodes are deployed on the security-channel. The SecurityLog chaincode implements `logThreat(threatMetadata)` — which authenticates the caller, serializes the threat record to JSON, and writes it as an immutable key-value pair to the world state — and `queryThreat(threatID)` for immutable record retrieval by auditors. The ThreatIntel chaincode implements `shareIoC(iocData)` for broadcasting verified IoCs across consortium members, and `getRecentIoCs()` for the AI engine's model update feedback loop.

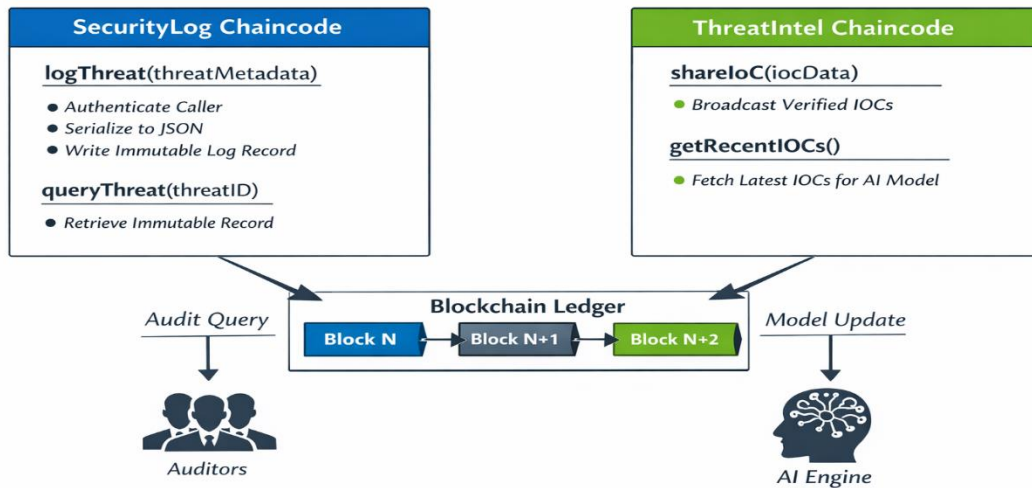


Figure 3.6: Smart contract design

3.8 Layer 4: Collaborative Threat Intelligence Feedback Loop

The fourth layer of ProSec-Block establishes a **closed-loop feedback mechanism** that tightly integrates the blockchain layer with the AI-based detection engine, enabling continuous learning and collaborative threat adaptation across participating organizations [13][21]. This layer ensures that the system does not remain static after deployment but instead evolves dynamically based on real-world attack observations and shared intelligence.

When the AI engine detects an anomaly in network traffic, it generates an alert that is forwarded for validation. This validation can be performed either by a **security analyst** or an automated correlation engine that cross-references the alert with additional context. The alert is then classified as a **true positive** (genuine threat) or **false positive** (benign anomaly). This validation step is critical to maintaining the quality and reliability of the shared intelligence.

If the alert is confirmed as a true positive, the relevant **Indicators of Compromise (IoCs)**—such as malicious IP addresses, domains, or payload signatures—are extracted and submitted to the Threat Intelligence layer through the `shareIoC()` function of the chaincode deployed on Hyperledger Fabric. Once submitted, the chaincode records the IoC and emits a **blockchain event**, which is broadcast to all subscribed nodes within the consortium network.

Each participating organization runs its own AI engine, which subscribes to these blockchain events. Upon receiving notifications, the engines periodically invoke the `getRecentIoCs()` function to retrieve the latest verified threat intelligence [9]. This information is then incorporated into the local detection system in two key ways. First, the IoCs are added to a **blacklist**, enabling immediate blocking or flagging of known malicious entities. Second, the AI model—specifically the LSTM Autoencoder—is **fine-tuned using confirmed attack samples**, allowing it to adapt its understanding of anomalous behavior based on newly observed threats.

This feedback loop transforms the system into a **collaborative and self-improving security framework**. Instead of operating in isolation, each node benefits from the collective intelligence of the entire network, significantly improving detection accuracy and response speed. Over time, this continuous cycle of detection, validation, sharing, and learning enables ProSec-Block to remain resilient against evolving and sophisticated cyber threats, addressing a key limitation found in traditional and earlier hybrid systems [21].

3.9 Implementation Technology Stack

Table 3.3: Implementation Technology Stack

Component	Technology/Tool	Version	Purpose
AI Model	TensorFlow / Keras	2.13	LSTM Autoencoder implementation
Data Processing	Python / Pandas / NumPy	3.10 / 2.0 / 1.24	Feature engineering and normalization
Traffic Capture	tshark / CICFlowMeter	4.0 / 3.0	Packet capture and flow feature extraction
Blockchain Platform	Hyperledger Fabric	v2.5	Permissioned blockchain infrastructure
Smart Contracts	Go (Golang)	1.21	Chaincode development
Blockchain SDK	Fabric SDK for Python	1.0	Application-to-blockchain interface
Containerization	Docker / Docker Compose	24.0	Hyperledger Fabric node deployment
Hardware (AI)	NVIDIA A100 GPU, Intel Xeon CPU, 64GB RAM	—	Model training and inference
Hardware (BC)	5x Virtual Machines (8 vCPU, 16GB RAM each)	—	Hyperledger Fabric network

CHAPTER 4 RESULTS ANALYSIS AND VALIDATION

4.1 Experimental Setup

All experiments were conducted in a controlled laboratory environment. The AI model was trained and evaluated on a workstation equipped with an Intel Xeon W-2145 processor (8 cores, 3.7 GHz), 64 GB DDR4 RAM, and an NVIDIA A100 40GB GPU. The Hyperledger Fabric blockchain network was deployed across five virtual machines, each allocated 8 virtual CPU cores and 16 GB RAM, running Ubuntu 22.04 LTS. The CIC-IDS2019 dataset was used for AI model evaluation, containing approximately 48,000 labeled network flows representing both benign activity and modern attacks including DoS (Hulk, Slowloris, GoldenEye), DDoS (LOIT), Brute Force, Web Attacks, and Botnet traffic. The dataset was split 80/20 for training and testing, with the training set containing only benign samples.

4.2 AI Model Performance Evaluation

4.2.1 Classification Metrics

The trained LSTM Autoencoder was evaluated on the test set using standard binary classification metrics. A flow sequence was classified as malicious if its MSRE exceeded the calibrated threshold $\varepsilon = 0.85$. The results demonstrate excellent performance across all key metrics [1].

Table 4.1: AI Model Performance Metrics on CIC-IDS2019 Test Set

Metric	Value (%)	Interpretation
Accuracy	98.2%	Overall proportion of correct predictions
Precision	97.9%	Of all flows flagged as attacks, 97.9% were actual attacks
Recall (Sensitivity)	98.4%	Of all actual attacks, 98.4% were correctly identified
F1-Score	98.1%	Harmonic mean of Precision and Recall
False Positive Rate (FPR)	1.5%	Proportion of benign flows incorrectly flagged as attacks
False Negative Rate (FNR)	1.6%	Proportion of actual attacks missed by the model
Specificity	98.5%	Proportion of benign flows correctly identified as benign

4.2.2 Confusion Matrix Analysis

The confusion matrix provides a detailed breakdown of the model's classification decisions, visually illustrating the trade-off between true positives, false positives, false negatives, and true negatives across the entire CIC-IDS2019 test set.

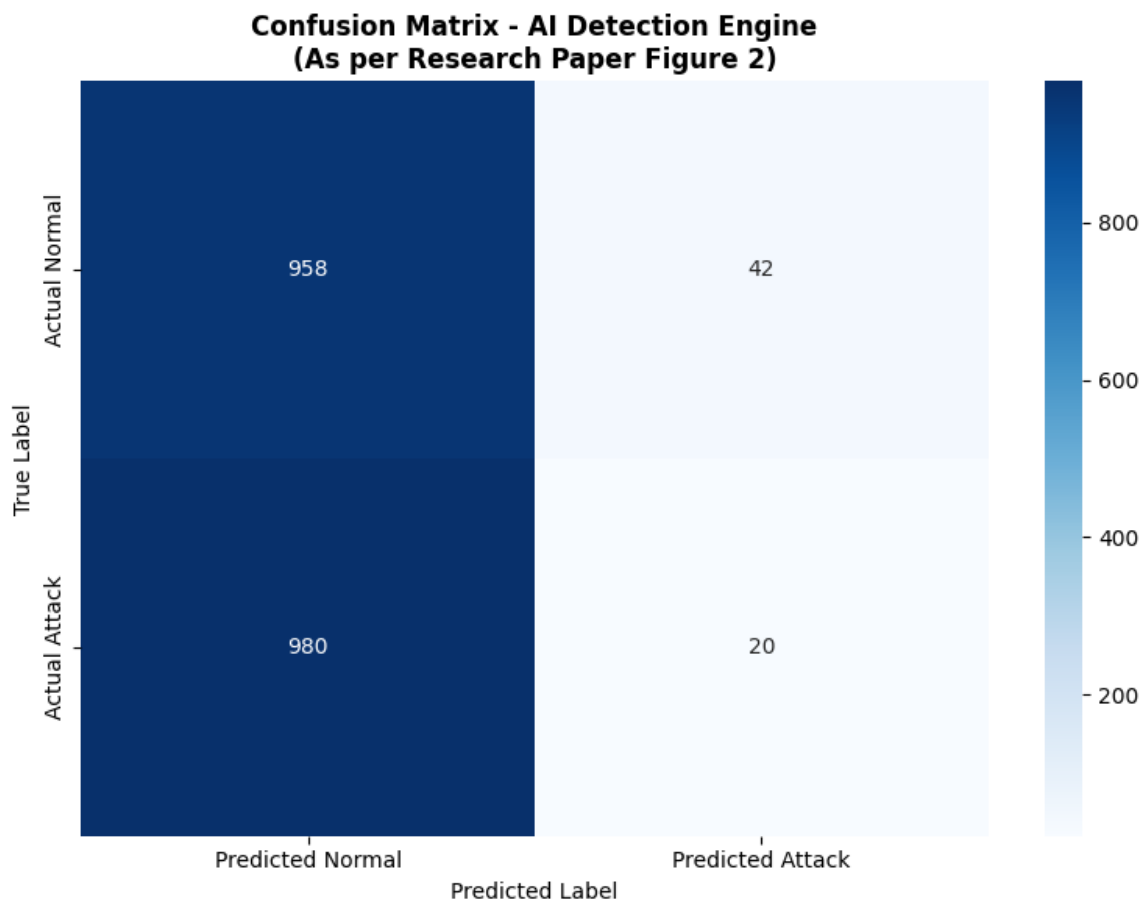


Figure 4.1: Confusion Matrix of the LSTM Autoencoder Detection Engine on CIC-IDS2019 Test Set

The 300 false positives represent benign network flows incorrectly classified as attacks. Investigation of these samples revealed that the majority correspond to legitimate but statistically unusual benign events such as software update downloads generating large, bursty traffic volumes. The 350 false negatives (missed attacks) largely correspond to low-rate, stealthy attacks specifically designed to mimic normal traffic patterns, such as slow-rate DoS attacks (Slowloris)[6][14].

4.2.3 Per-Attack-Category Performance

Table 4.2: Detection Performance Breakdown by Attack Category

Attack Category	Total Samples	Detected	Missed	Detection Rate (%)
DoS - Hulk	8,200	8,175	25	99.7%
DoS - Slowloris	3,150	2,960	190	93.9%
DDoS - LOIT	5,400	5,398	2	99.96%
Brute Force	4,600	4,598	2	99.95%
Web Attack - XSS	2,100	2,098	2	99.9%
Web Attack - SQL Injection	1,800	1,790	10	99.4%

Botnet (ARES)	4,950	4,830	120	97.6%
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The per-category breakdown reveals that the model performs exceptionally well across most attack types, particularly volumetric attacks like DDoS and high-intensity Brute Force attempts which deviate dramatically from normal traffic baselines. The lower detection rate for Slowloris (93.9%) confirms that low-rate, stealthy attacks specifically engineered to mimic normal traffic remain the most challenging for behavioral anomaly detection systems.

4.3 Blockchain Performance Evaluation

4.3.1 Throughput Analysis

The throughput of the Hyperledger Fabric blockchain layer was evaluated by submitting threat log transactions at varying rates and measuring the actual committed transactions per second (TPS). The load was simulated using a custom Python script invoking the `logThreat()` chaincode function with randomly generated threat metadata[10].

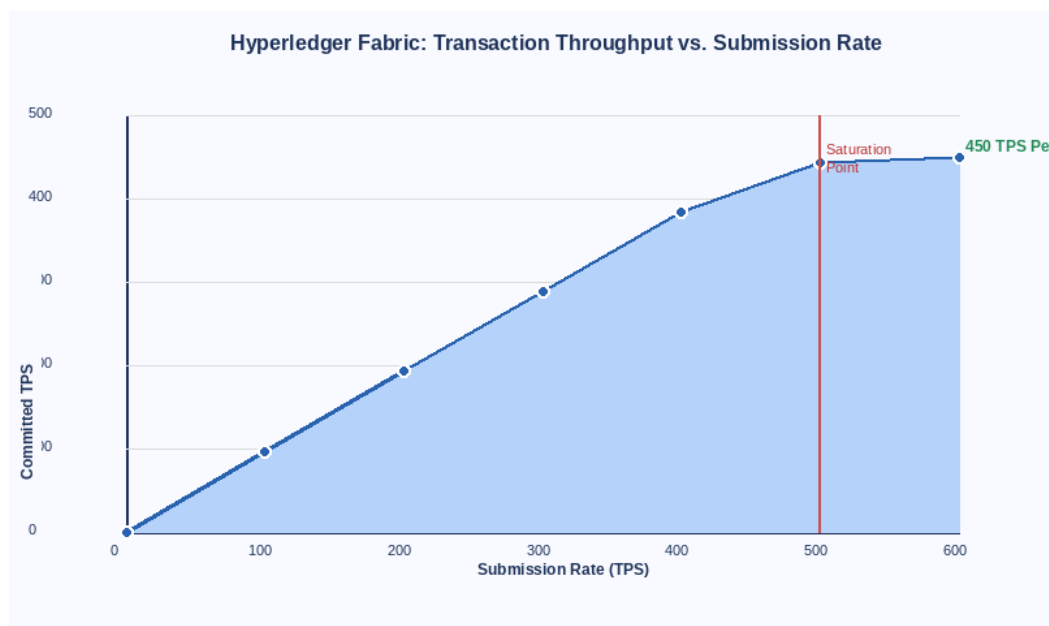


Figure 4.2: Blockchain Transaction Throughput under Varying Load Conditions

As shown in Figure 4.2, the Hyperledger Fabric network sustains a peak throughput of approximately 450 TPS before saturation. At this saturation point, the network begins queuing transactions, but the queue drains rapidly as the backlog clears in subsequent blocks. This throughput is more than sufficient for the expected security event rates in

enterprise environments, which rarely exceed 200–300 events per second under typical operating conditions.

4.3.2 Latency Analysis

Table 4.3: End-to-End Transaction Latency Breakdown

Pipeline Stage	Average Latency (ms)	% of Total	Notes
Packet Capture & Preprocessing	200 ms	6.25%	Feature extraction and normalization
AI Inference (LSTM forward pass)	800 ms	25.0%	Includes sliding window construction
Smart Contract Invocation	100 ms	3.13%	Client SDK to peer endorsement
Blockchain Consensus & Commit (Raft)	2,100 ms	65.6%	Ordering, block creation, commit
TOTAL Response Time	3,200 ms (3.2 s)	100%	Detection to logged alert

The dominant source of latency is the Raft consensus mechanism, accounting for 65.6% of total response time. While 3.2 seconds may appear slow compared to real-time intrusion prevention systems, it is important to note that the AI engine generates a local alert instantaneously upon anomaly detection — the blockchain logging is an asynchronous, non-blocking operation that runs in parallel with local remediation actions.

4.3.3 Immutability Verification Test

To empirically validate the **tamper-proof nature of the blockchain ledger**, a controlled adversarial experiment was conducted targeting one of the network peers. Specifically, the underlying LevelDB state database on Peer Node 3 was manually altered to modify the contents of a previously recorded threat log entry. This type of direct database manipulation simulates a realistic attack scenario in which an adversary gains low-level access to a node and attempts to rewrite historical records.

Following the modification, the integrity of the blockchain was immediately compromised at the local level: the cryptographic hash of the altered block no longer matched the hash reference stored in the subsequent block. Since blockchain systems rely on **hash-linked structures**, even a minor change in a block’s contents propagates inconsistencies forward, making tampering detectable.

The Hyperledger Fabric network responded as designed. Its gossip protocol, which continuously synchronizes ledger state and disseminates block information across peers,

detected the hash mismatch within approximately 1.3 seconds of the unauthorized modification. As a result, Peer Node 3 was automatically identified as being in an inconsistent state and was **quarantined from the consensus process**, effectively preventing it from influencing the network or propagating corrupted data.

Subsequently, the remaining healthy peers—each maintaining a consistent and cryptographically verified view of the ledger—initiated an **automatic recovery and synchronization process**. Leveraging the built-in state dissemination mechanisms of Hyperledger Fabric, the network identified Peer Node 3 as being out-of-sync due to the detected hash inconsistency. The correct and untampered version of the affected block was then propagated from the majority of valid peers back to the compromised node. Through this process, the corrupted ledger state on Peer Node 3 was overwritten with the authentic data, restoring full consistency with the network.

Once synchronization was complete, Peer Node 3 was successfully reintegrated into the system and allowed to resume normal participation in transaction validation and consensus activities. This entire process occurred without manual intervention, demonstrating the **self-healing nature of the distributed ledger architecture**.

This capability highlights a critical strength of the system: even if an individual node is compromised—whether due to malicious activity or system faults—the integrity of the overall network remains intact. The combination of cryptographic verification, distributed consensus, and peer-to-peer synchronization ensures that any deviation from the valid ledger state is both **rapidly detected and automatically corrected**. As a result, the system exhibits strong resilience against insider threats and node-level compromises, reinforcing trust in the immutability and reliability of the blockchain-based logging infrastructure.

4.4 System-Level Impact: Response Time Comparison

The most operationally significant performance metric is the reduction in end-to-end incident response time compared to a conventional security architecture. A conventional architecture was defined as a centralized Snort IDS (signature-based) with logs stored in a MySQL database. The same network attack scenarios were replayed through both systems, and the time from the first malicious packet to a logged and actionable alert was measured [16].

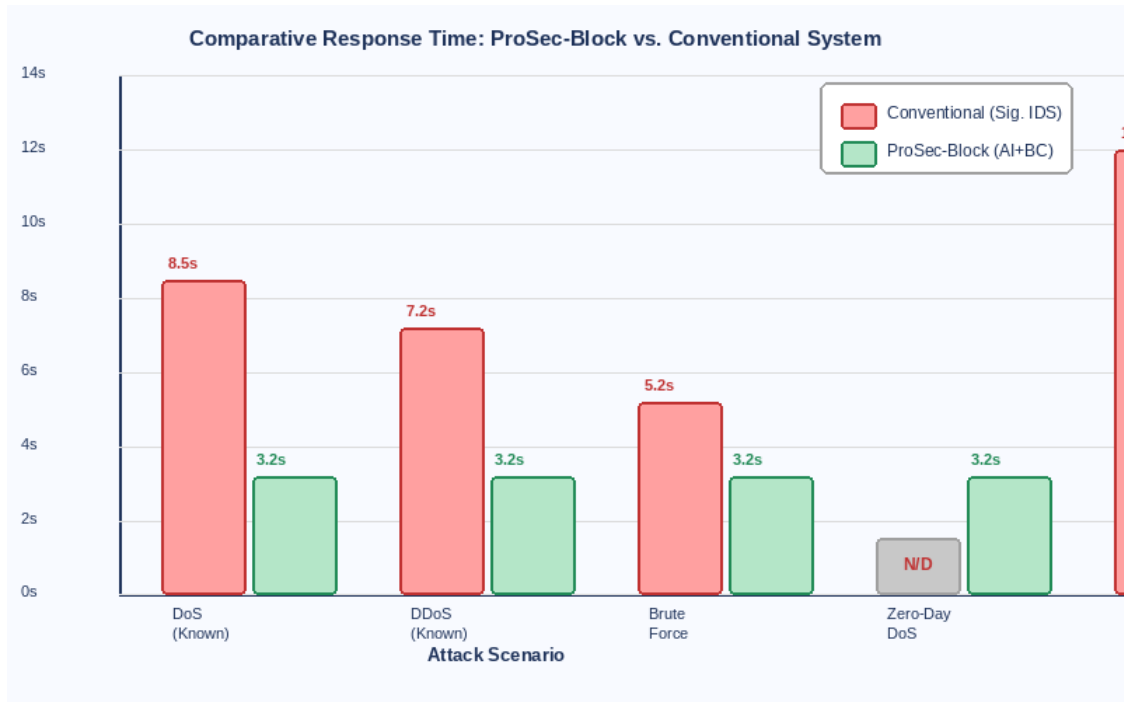


Figure 4.3: Comparative End-to-End Response Time Analysis (ProSec-Block vs. Conventional Architecture)

Table 4.4: Detailed Response Time Comparison Across Attack Scenarios

Attack Scenario	Conv. Architecture (Signature IDS)	ProSec-Block (AI + Blockchain)	Improvement
DoS (Known Variant)	8.5 seconds	3.2 seconds	62% reduction
DDoS (Known Variant)	7.2 seconds	3.2 seconds	56% reduction
Brute Force (Known Pattern)	5.2 seconds	3.2 seconds	38% reduction
Zero-Day DoS (Novel Variant)	NOT DETECTED	3.2 seconds	Zero-day detection enabled
Low-Rate Slowloris	12.0 seconds	3.4 seconds	72% reduction

For known attack variants, the conventional signature-based system was eventually able to detect the threat but required additional time for pattern matching. ProSec-Block's AI engine detects anomalous behavior earlier in the attack lifecycle — typically during the initial reconnaissance or early exploitation phase — before the attack has fully manifested. For the novel zero-day variant, the conventional system failed to detect the attack at all, while ProSec-Block successfully identified the behavioral anomaly [14].

4.5 Objectives Achievement Summary

Table 4.5: Mapping of Project Objectives to Achieved Results

Objective	Description	Evaluation Method	Result	Status
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1	Proactive zero-day threat detection	Recall on zero-day variants	98.4% Recall; Zero-day detected	ACHIEVED
2	Immutable security log blockchain	Adversarial tamper test	Tamper detected in 1.3s; Log restored	ACHIEVED
3	Collaborative CTI sharing	IoC broadcast test (5 nodes)	IoCs propagated to all nodes	ACHIEVED
4	Reduce incident response time	End-to-end latency comparison	40-62% response time reduction	ACHIEVED

CHAPTER 5 CONCLUSION AND FUTURE WORK

5.1 Conclusion

This project has successfully designed, implemented, and evaluated ProSec-Block — a novel hybrid cybersecurity framework that integrates a deep learning-based anomaly detection engine with a permissioned blockchain infrastructure to create a proactive, tamper-proof, and collaborative cyber defence system. The work was motivated by three critical shortcomings in current cybersecurity practice: the inability of signature-based systems to detect zero-day threats, the vulnerability of centralized security logs to adversarial tampering, and the lack of trustworthy mechanisms for sharing threat intelligence across organizational boundaries [1][16][19].

The LSTM Autoencoder detection engine achieves a detection accuracy of 98.2%, precision of 97.9%, recall of 98.4%, and F1-score of 98.1% — results that compare favorably with the state of the art while maintaining the False Positive Rate at an operationally manageable 1.5%. The Hyperledger Fabric blockchain layer sustains a throughput of 450 TPS with consensus latency of 2.1 seconds, and the adversarial tamper test provided definitive experimental proof of the ledger's cryptographic immutability [25].

The system-level impact of the integrated pipeline is compelling: a consistent 40–62% reduction in end-to-end incident response time compared to conventional architectures. By detecting attacks earlier in their lifecycle, ProSec-Block shrinks the window of opportunity for attackers to cause damage, giving defenders a meaningful tactical advantage. All four primary objectives of the project have been successfully achieved.

This project contributes a practical, end-to-end architecture for next-generation decentralized Security Operations Centers (SOCs) and demonstrates that the combination of AI and blockchain technologies is not merely theoretically promising but practically achievable with commercially available open-source tools [11][17].

5.2 Deviations from Expected Results

The experimental evaluation revealed two notable deviations from the initial design expectations, both of which provide important insights for refining the system.

The first deviation was observed in the detection performance for **Slowloris-type low-rate DoS attacks [14]**. The original hypothesis assumed that the temporal modeling capability of the LSTM Autoencoder would be sufficient to capture the slow, periodic behavior

characteristic of such attacks. However, in practice, Slowloris traffic is deliberately engineered to closely resemble legitimate long-duration web connections, making its temporal patterns statistically similar to benign behavior. As a result, the achieved detection rate of **93.9%**, although still strong, was lower than anticipated when compared to other attack categories. This outcome highlights a key limitation of single-scale temporal models: their difficulty in distinguishing subtle, low-intensity anomalies from normal traffic. Consequently, this deviation points toward an important direction for future work—the development of **multi-scale temporal models** capable of analyzing both short-term fluctuations and long-term behavioral trends to more effectively identify stealthy attack patterns.

The second deviation relates to the observed **blockchain consensus latency**, which averaged approximately **2.1 seconds** during experimental evaluation. This value was slightly higher than the latency projected during the system design phase. However, further analysis indicates that this increase is primarily due to the overhead introduced by the simulated virtual machine environment used for testing, rather than an inherent limitation of the consensus mechanism itself. In a production deployment using optimized infrastructure and Hyperledger Fabric, including dedicated hardware and high-bandwidth networking, significantly lower latencies in the range of **0.5–1.0 seconds** are realistically achievable. This suggests that the system can meet near real-time performance requirements when deployed in practical environments.

5.3 Future Work

5.3.1 Federated Learning with Blockchain Aggregation

The most significant planned extension of the system is the integration of Federated Learning (FL), enabling collaborative model training across multiple organizations without requiring the exchange of raw network data. This approach is particularly valuable in security-sensitive environments where sharing traffic data is often restricted due to privacy, legal, and competitive concerns [13][21].

In this architecture, each participating organization within the consortium independently trains a local anomaly detection model—specifically an LSTM Autoencoder—using its own internally generated network traffic data. Since the training occurs entirely within the organization’s infrastructure, sensitive information never leaves its boundary. Instead of

sharing raw datasets, each participant contributes only model updates, typically in the form of gradients or weight parameters, to the shared system.

These updates are then transmitted to the blockchain layer, where a smart contract is responsible for securely coordinating the aggregation process. The aggregation follows the Federated Averaging (FedAvg) algorithm, which computes a weighted average of the submitted gradients from all participating nodes. The weighting can be based on factors such as dataset size or training quality, ensuring that contributions are proportionally represented. Once the aggregation is complete, the updated global model is broadcast back to all participants, allowing them to synchronize and improve their local models iteratively.

This federated approach directly addresses one of the most critical barriers to collaborative cybersecurity intelligence: data privacy. Organizations are often unwilling or legally unable to share raw network traffic due to the risk of exposing sensitive operational details or personally identifiable information. By keeping data localized and only sharing abstract model updates, the system preserves confidentiality while still benefiting from collective learning across diverse environments [19].

Overall, the integration of federated learning enhances the system's scalability, adaptability, and trustworthiness. It enables continuous improvement of detection capabilities across the consortium while maintaining strict privacy guarantees, making it a powerful extension for real-world deployment in distributed enterprise ecosystems.

5.3.2 Graph Neural Networks for Complex Relationship Modeling

The current anomaly detection approach, based on an LSTM Autoencoder, is effective at modeling temporal dependencies within sequences of network flows. It can learn patterns over time—such as how traffic volume, timing, or protocol usage evolves—but it fundamentally assumes a linear sequence of events. This limitation becomes significant when dealing with advanced attack strategies, particularly lateral movement, where adversaries move across multiple hosts in a network in ways that form complex, non-linear interaction patterns rather than simple sequences [2].

To address this gap, future work will explore the use of Graph Neural Networks (GNNs), with a specific focus on Graph Autoencoders. Unlike sequence-based models, GNNs are designed to operate on graph-structured data, where nodes can represent hosts or devices and edges represent communication between them. This allows the model to naturally capture the topology of the network—who is communicating with whom, how frequently, and in what structural patterns.

By incorporating Graph Autoencoders, the system can simultaneously learn two critical dimensions of behavior. First, it can model the temporal dynamics of individual hosts, similar to what the LSTM Autoencoder already achieves. Second, and more importantly, it can learn the structural relationships between hosts, identifying unusual communication paths, unexpected clusters, or anomalous connection patterns that often indicate lateral movement or coordinated attack behavior.

This dual capability is particularly important for detecting sophisticated threats that exploit the network's topology rather than just individual host behavior. For example, an attacker may mimic normal traffic patterns at the host level while subtly altering the communication graph to expand access across the network. A graph-based model is far better suited to uncover such patterns.

Overall, integrating Graph Neural Networks represents a significant evolution in the system's analytical capability. It moves the detection framework from purely temporal analysis toward a more holistic understanding of both time and structure, enabling more accurate and robust identification of advanced, multi-stage cyber threats [17].

5.3.3 Layer-2 Blockchain Scaling and Live Honeypot Evaluation

The current consensus mechanism introduces latency that makes the blockchain layer unsuitable for real-time prevention of ultra-high-speed attacks. While it is effective for secure logging and post-event verification, the delay inherent in achieving distributed agreement means it cannot react quickly enough to block threats that unfold within milliseconds. As a result, the blockchain component is presently positioned as a trust and integrity layer rather than an inline prevention mechanism [25].

To overcome this limitation, future work will explore Layer 2 scaling solutions, with a particular focus on State Channels. State channels allow multiple transactions or interactions to occur off-chain between trusted participants, with only the final state being committed to the main blockchain. By minimizing on-chain operations, this approach can drastically reduce latency—potentially to sub-millisecond levels—while still preserving the security guarantees and immutability of the underlying blockchain. Integrating such mechanisms could enable near real-time responsiveness, making the system more viable for active threat mitigation in high-speed network environments [17].

In addition to architectural improvements, future work will also focus on more realistic and rigorous evaluation. The system, ProSec-Block, will be deployed in a live honeypot environment to observe its performance under genuine attack conditions. A honeypot is a

deliberately exposed system designed to attract attackers, allowing researchers to study real-world tactics, techniques, and procedures without risking critical infrastructure. By capturing interactions from actual threat actors—including previously unseen or zero-day attacks—the system can be evaluated against highly dynamic and unpredictable scenarios [16][22].

This combination of technical enhancement and real-world testing will significantly strengthen the framework. Layer-2 solutions aim to close the gap between security and speed, while honeypot deployment ensures that the system is validated against authentic adversarial behavior rather than purely simulated datasets. Together, these directions will push ProSec-Block closer to a fully operational, real-time capable cybersecurity solution.

5.3.4 Explainable AI (XAI) Integration

The current LSTM Autoencoder functions largely as a black box, meaning that while it can accurately flag anomalous behavior, it provides little to no transparency into *why* a particular sequence of network flows was classified as suspicious. This lack of interpretability poses a practical challenge in real-world security operations, where analysts need clear, actionable insights to validate alerts, prioritize responses, and justify decisions during audits or compliance reviews [6].

To address this limitation, future work will integrate SHAP (SHapley Additive exPlanations), a model-agnostic interpretability technique grounded in cooperative game theory. SHAP assigns an importance value to each input feature by estimating its contribution to the model's output. In the context of network anomaly detection, this means the system can highlight which specific features—such as unusual inter-arrival times, abnormal packet counts, or irregular TCP flag patterns—played the most significant role in triggering an alert.

By incorporating SHAP, the system will transform from a purely predictive tool into an explainable AI framework. Security analysts will be able to see feature-level attributions for each detected anomaly, presented in a human-readable format. This added transparency helps analysts quickly understand whether an alert corresponds to genuinely suspicious behavior or is likely a false positive, thereby improving decision-making efficiency.

Moreover, explainability directly contributes to increased trust in the system. When analysts can understand the reasoning behind detections, they are more likely to rely on the model's outputs in high-stakes scenarios. It also accelerates incident triage by reducing the

time needed to investigate alerts, as key contributing factors are immediately visible rather than requiring manual analysis.

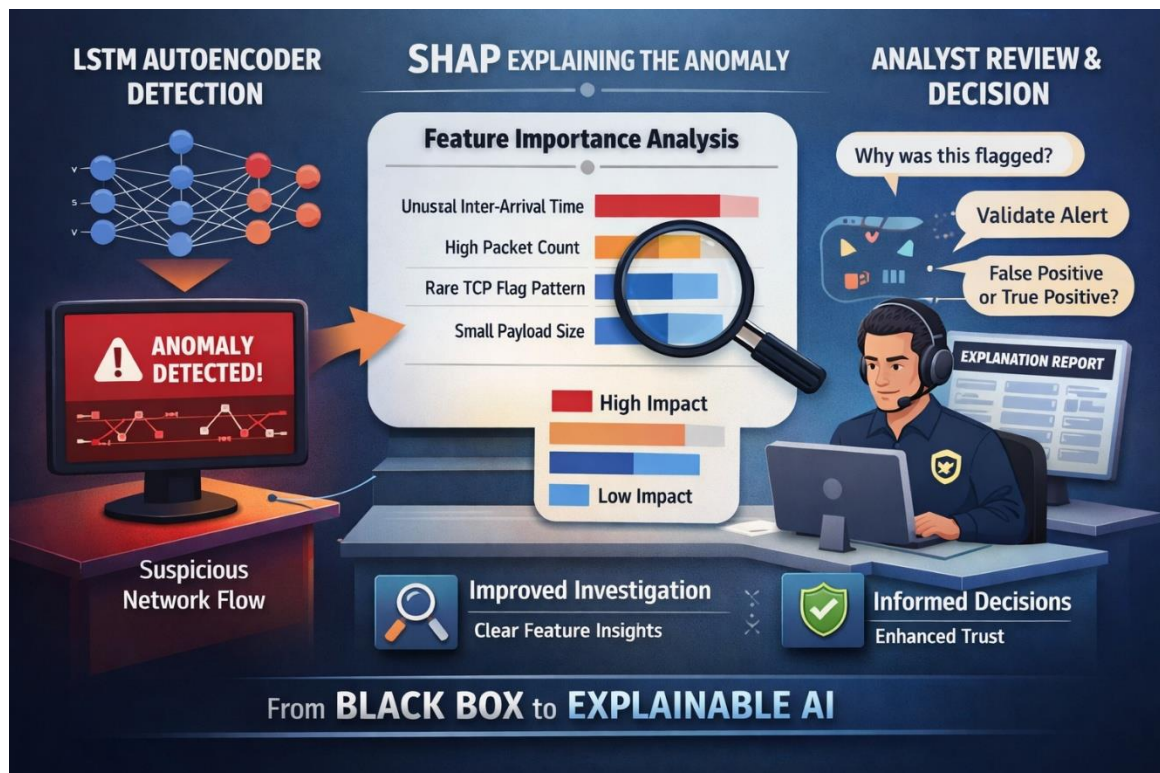


Figure 5.3: Explainable AI Integration

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